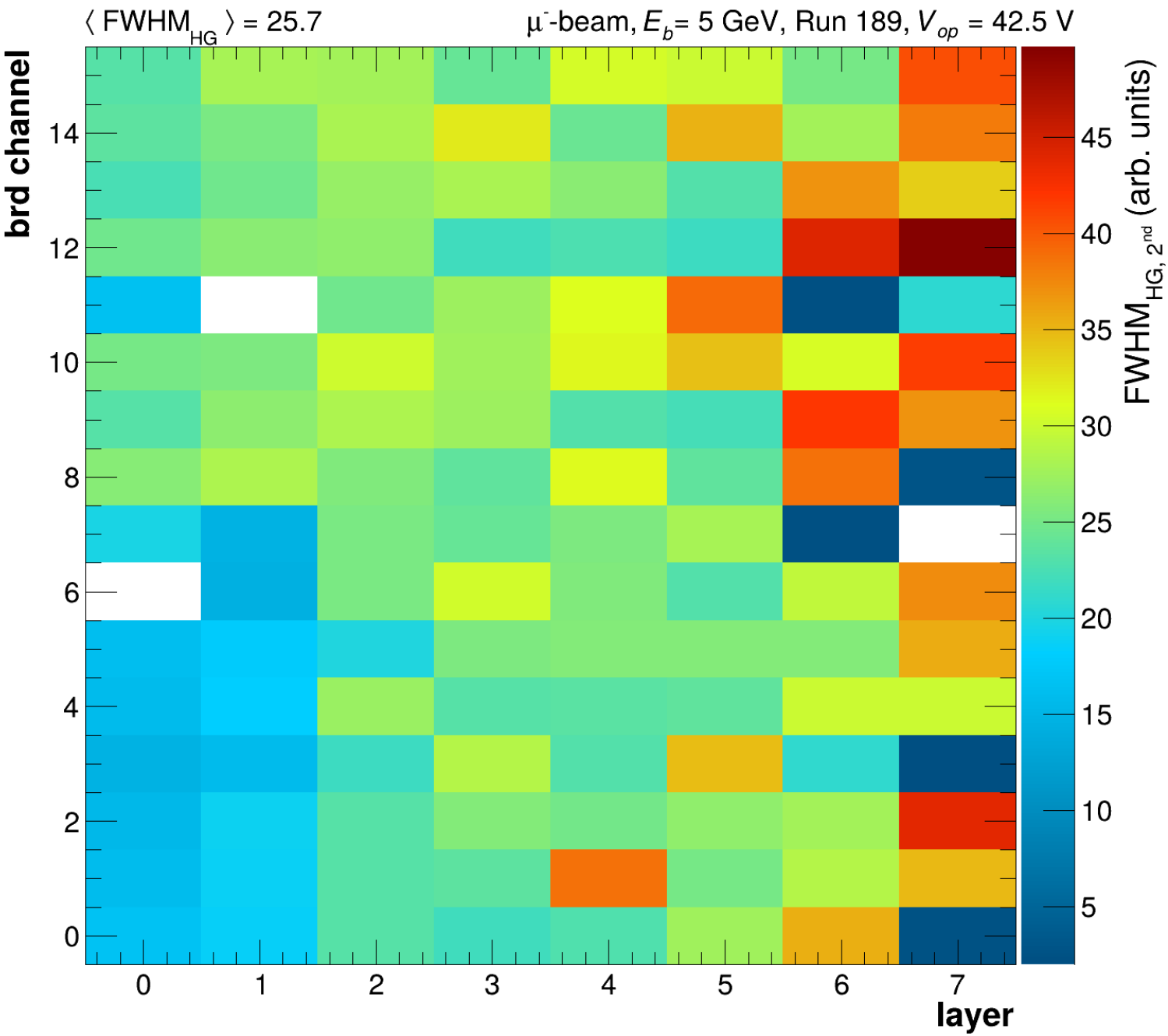
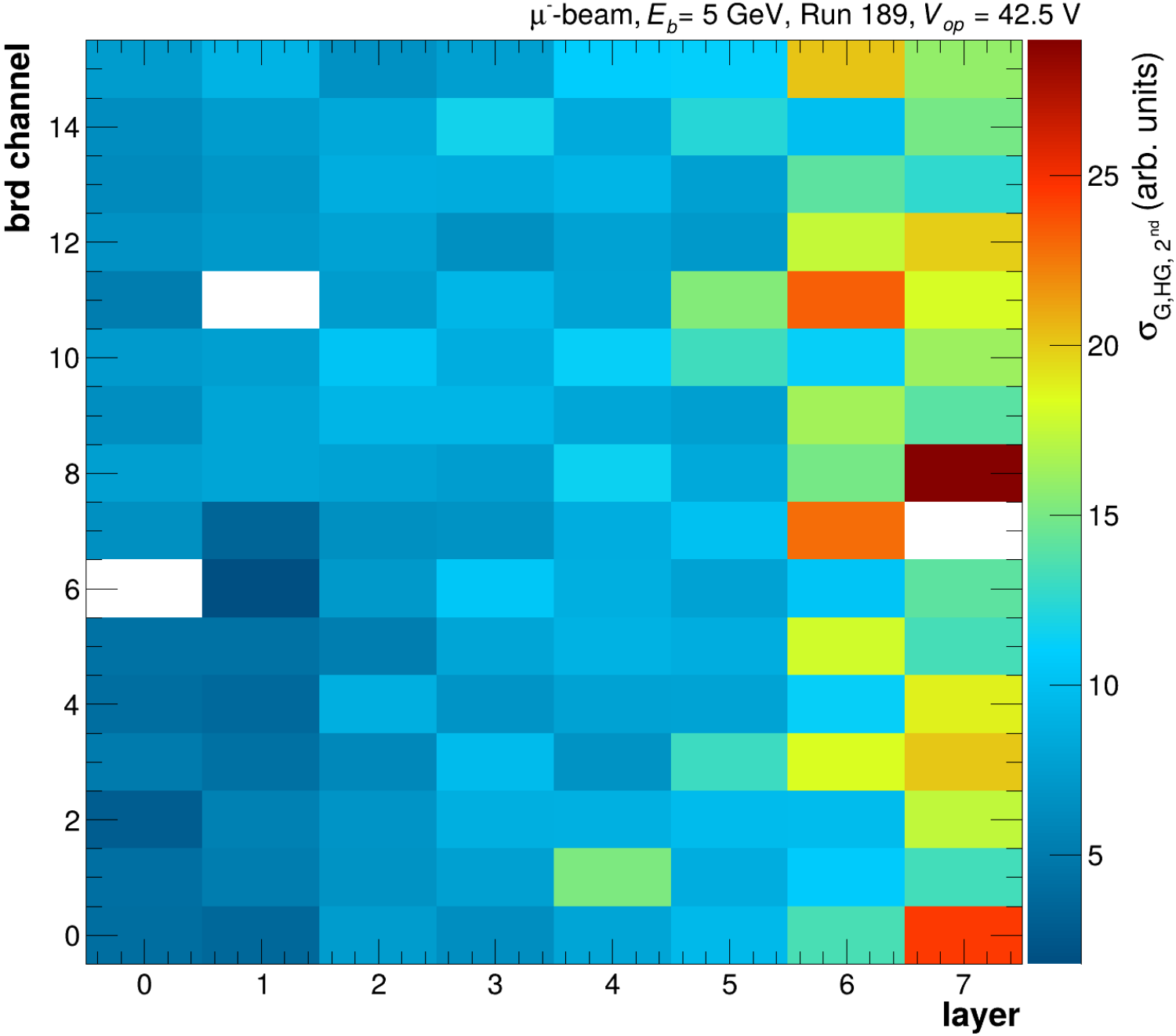
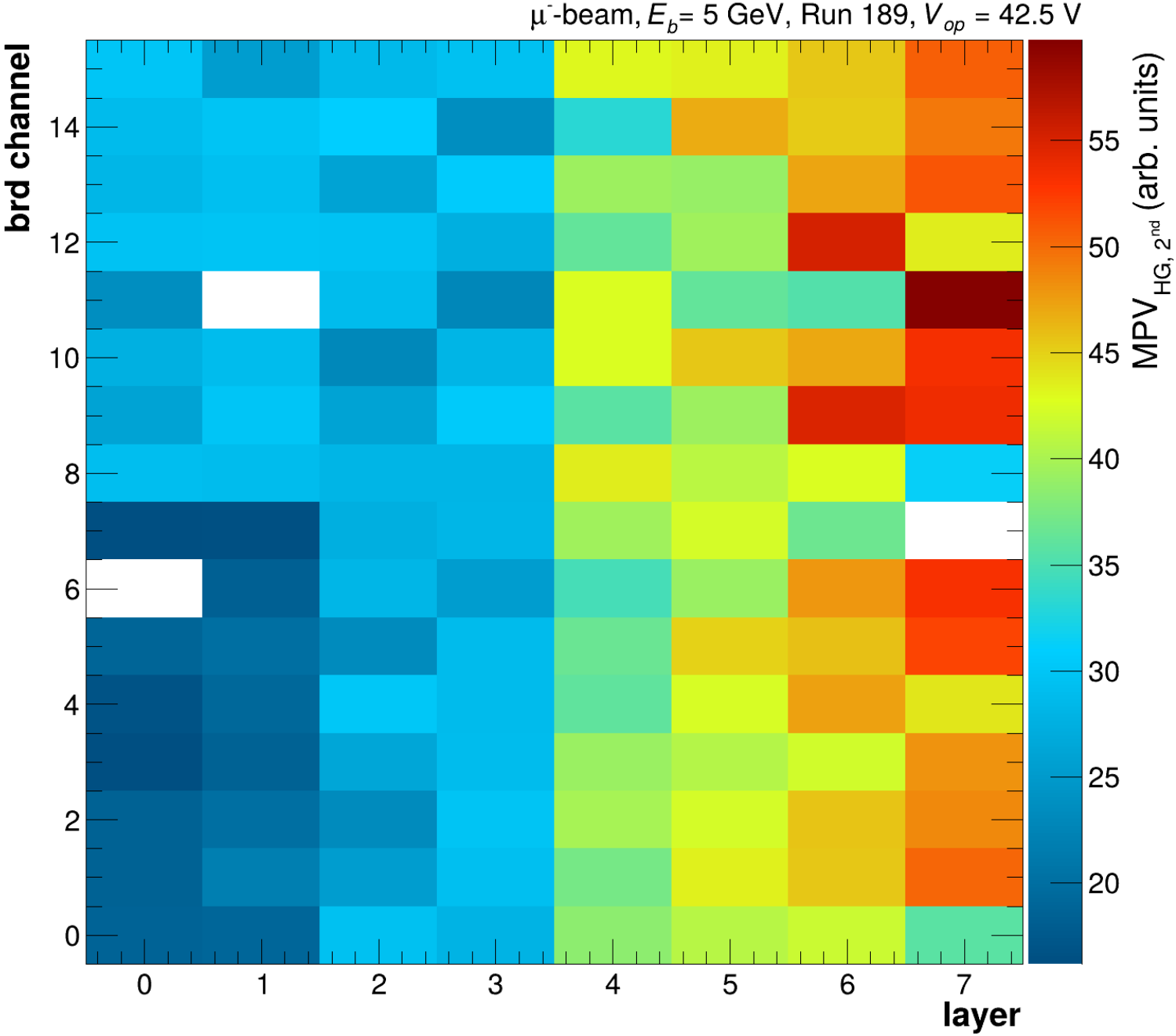
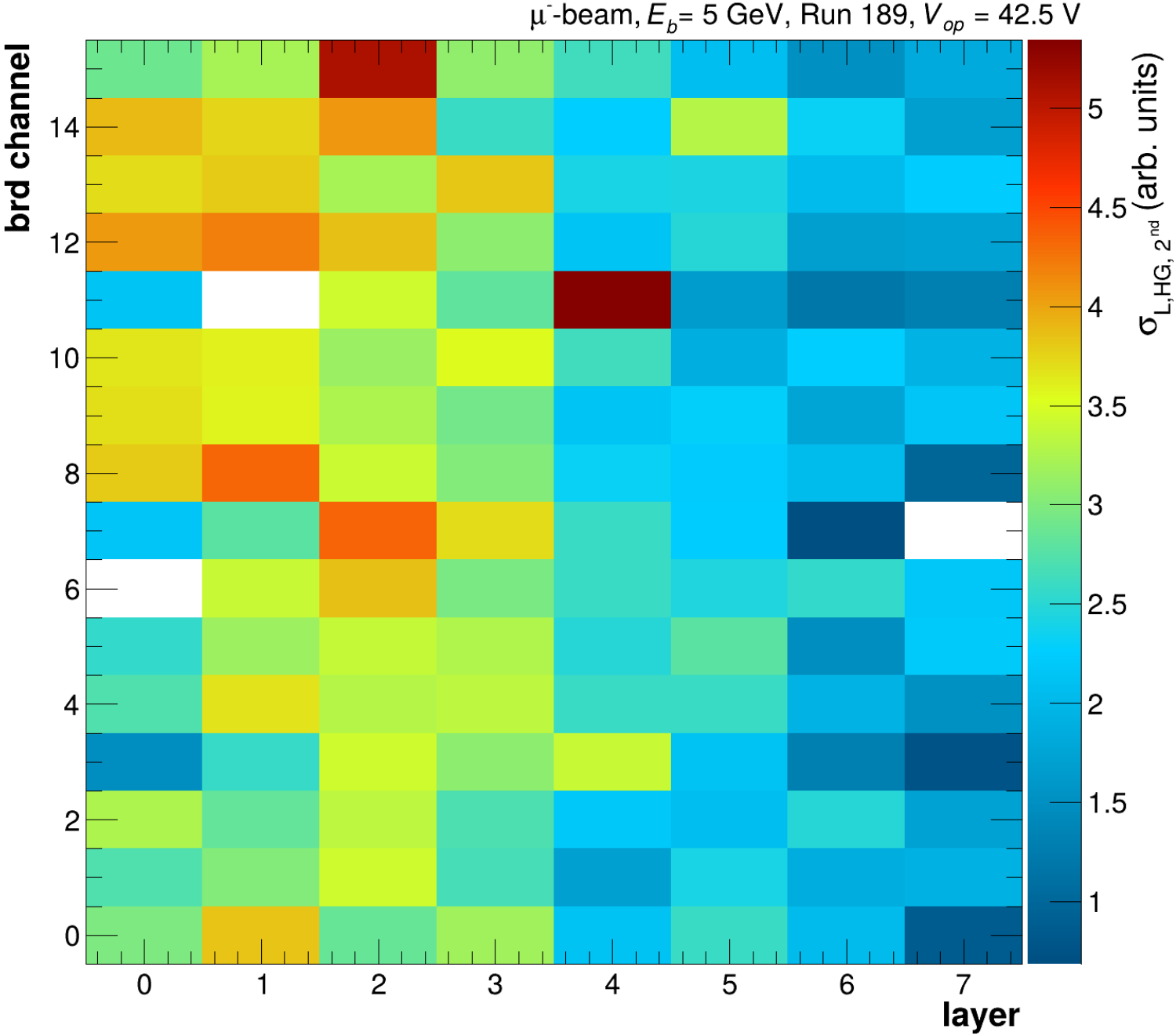
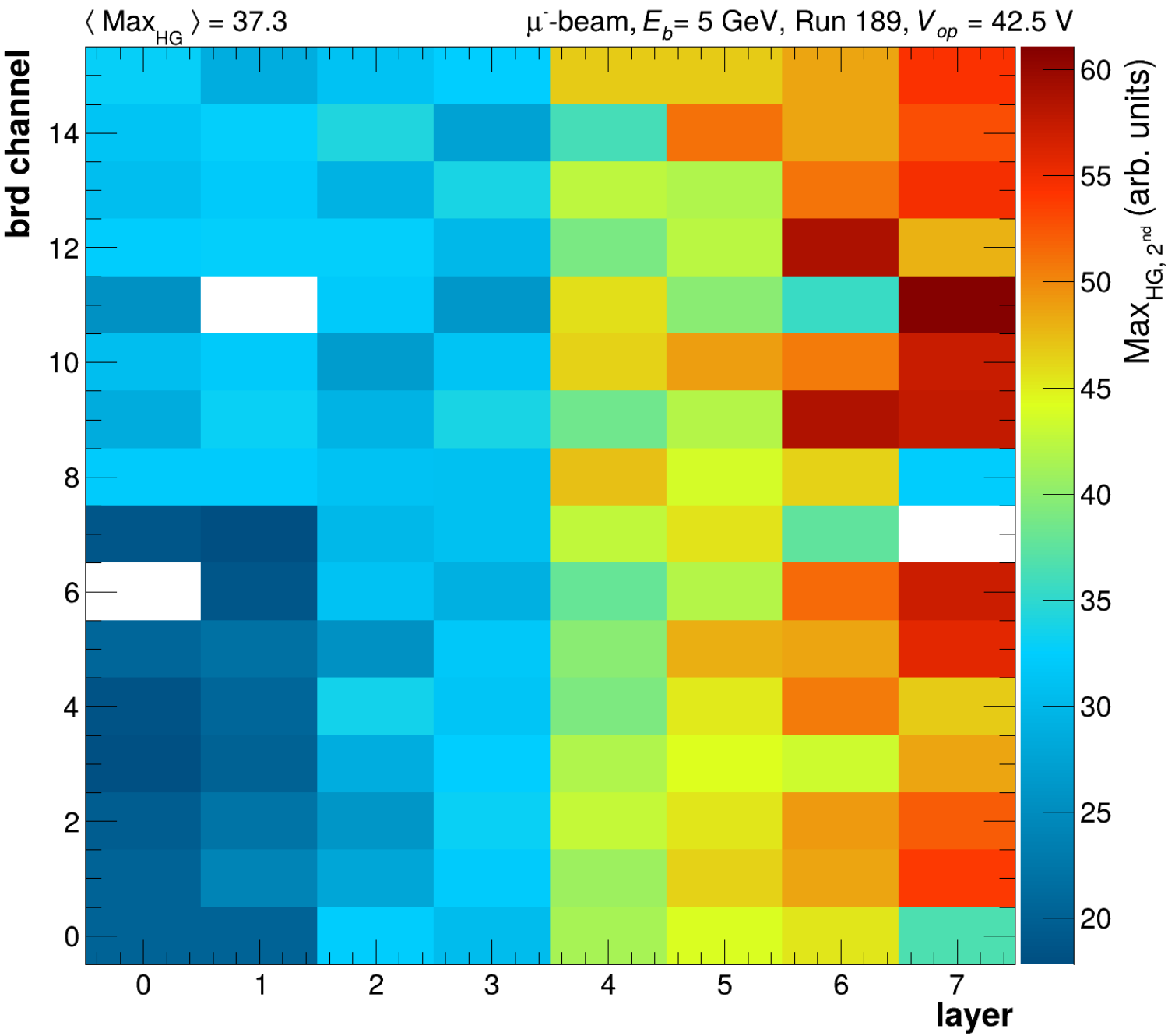
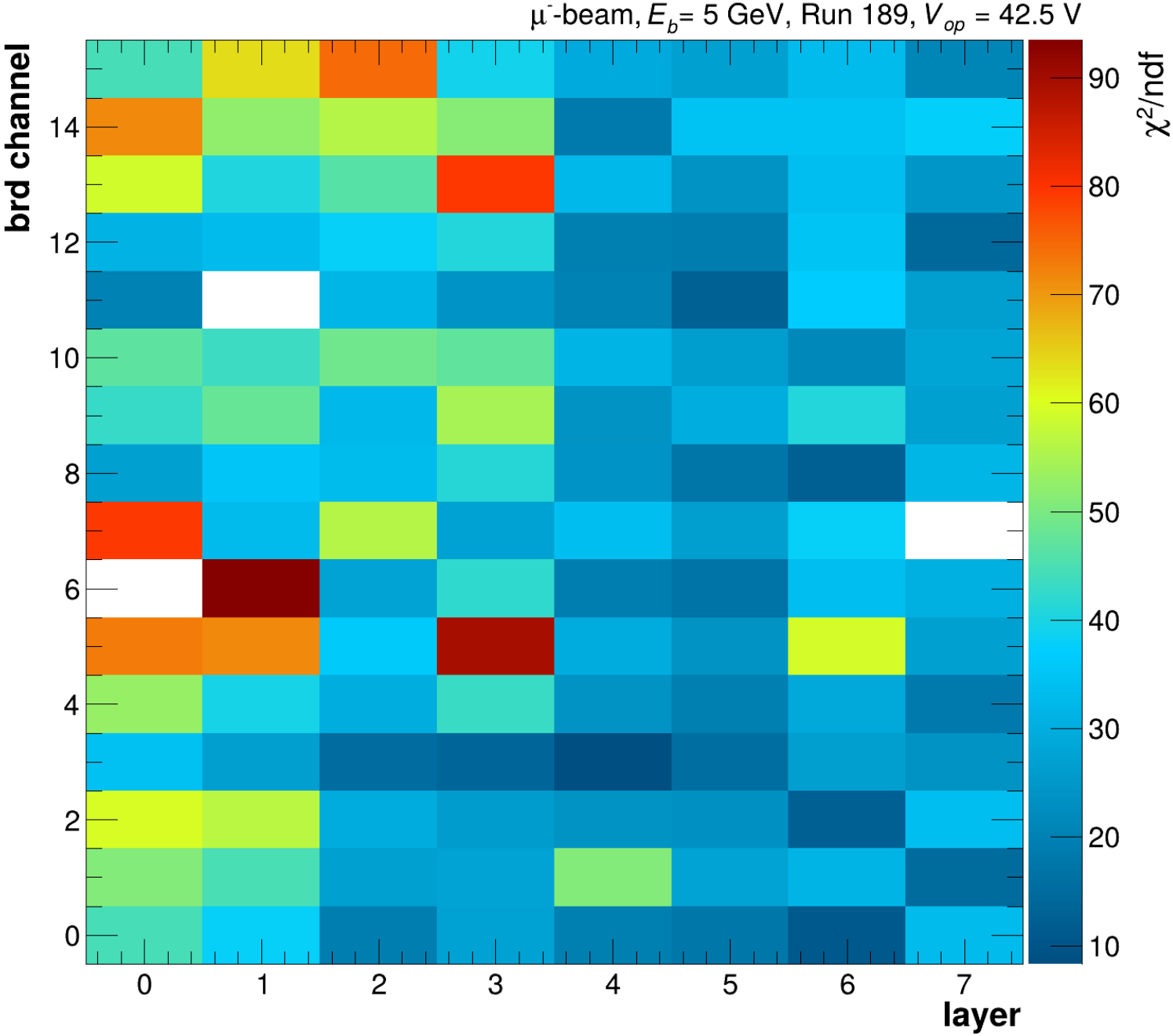

⟨ FWHM_HG ⟩ = 25.7
μ⁻-beam, E_b = 5 GeV, Run 189, V_op = 42.5 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
5
10
15
20
25
30
35
40
45

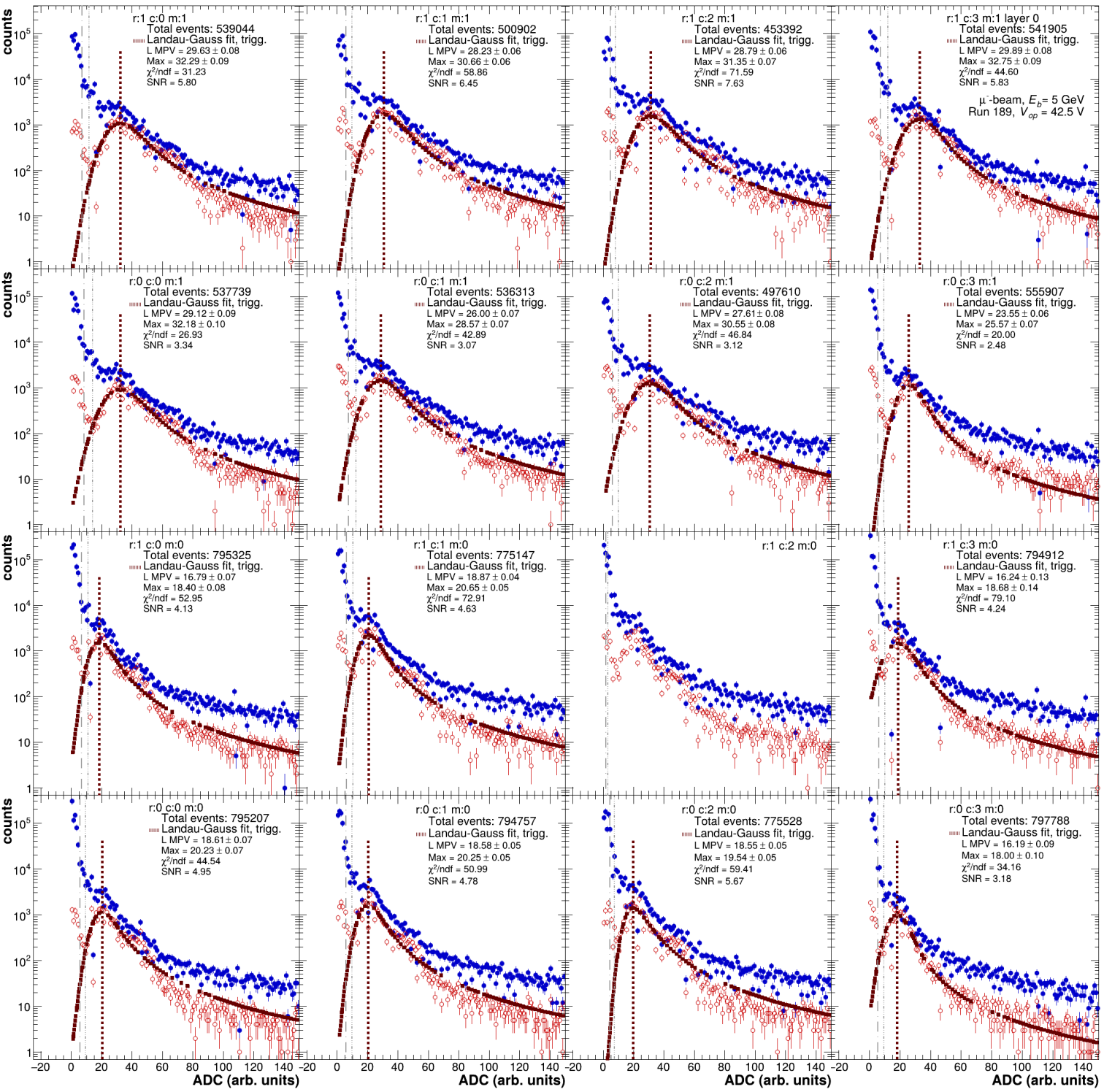

μ⁻-beam, E_b= 5 GeV, Run 189, V_{op} = 42.5 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
5
10
15
20
25

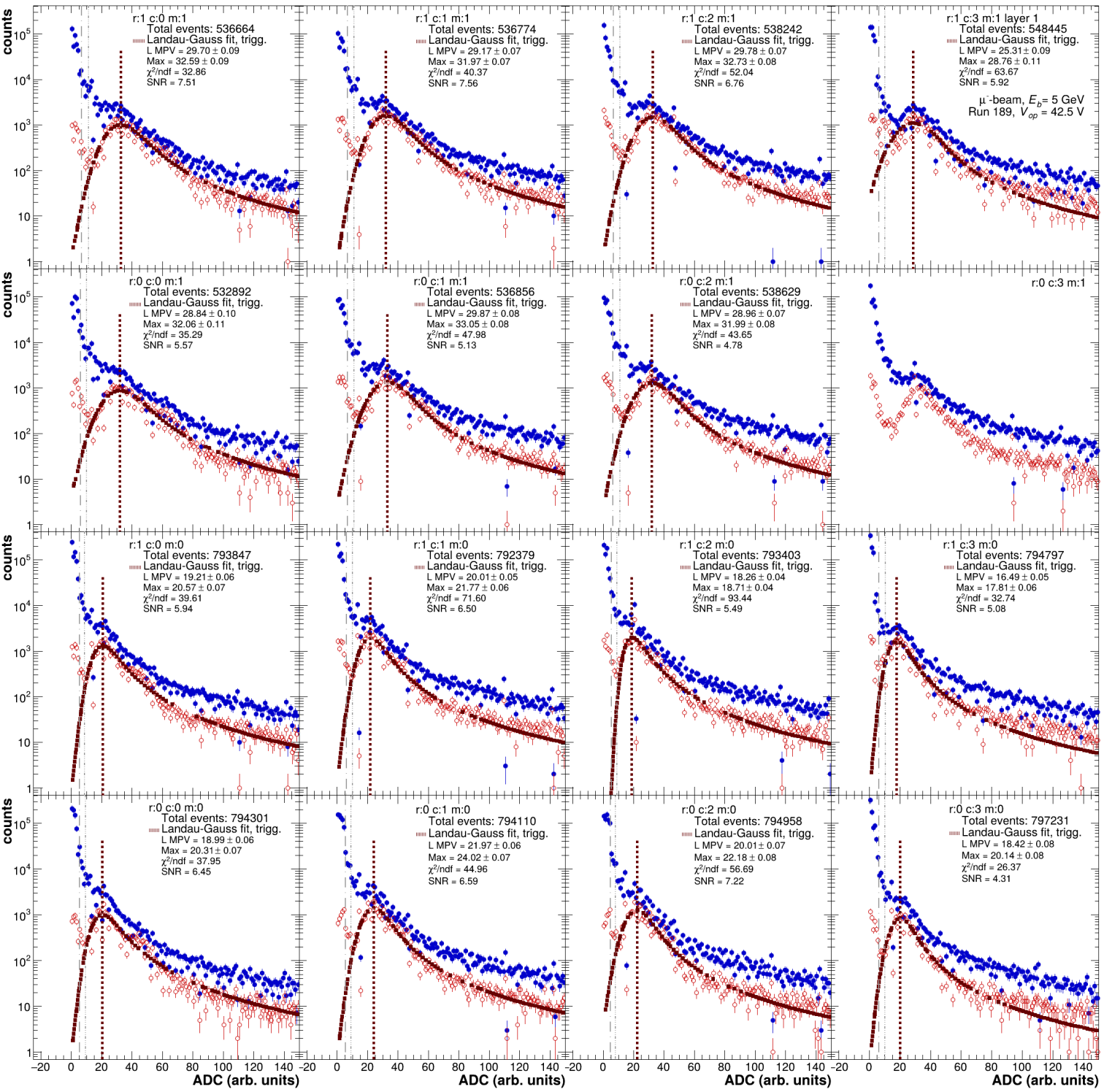

μ⁻-beam, E_b= 5 GeV, Run 189, V_{op} = 42.5 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$MPV_{HG,\ 2^{nd}}$ (arb. units)
20
25
30
35
40
45
50
55

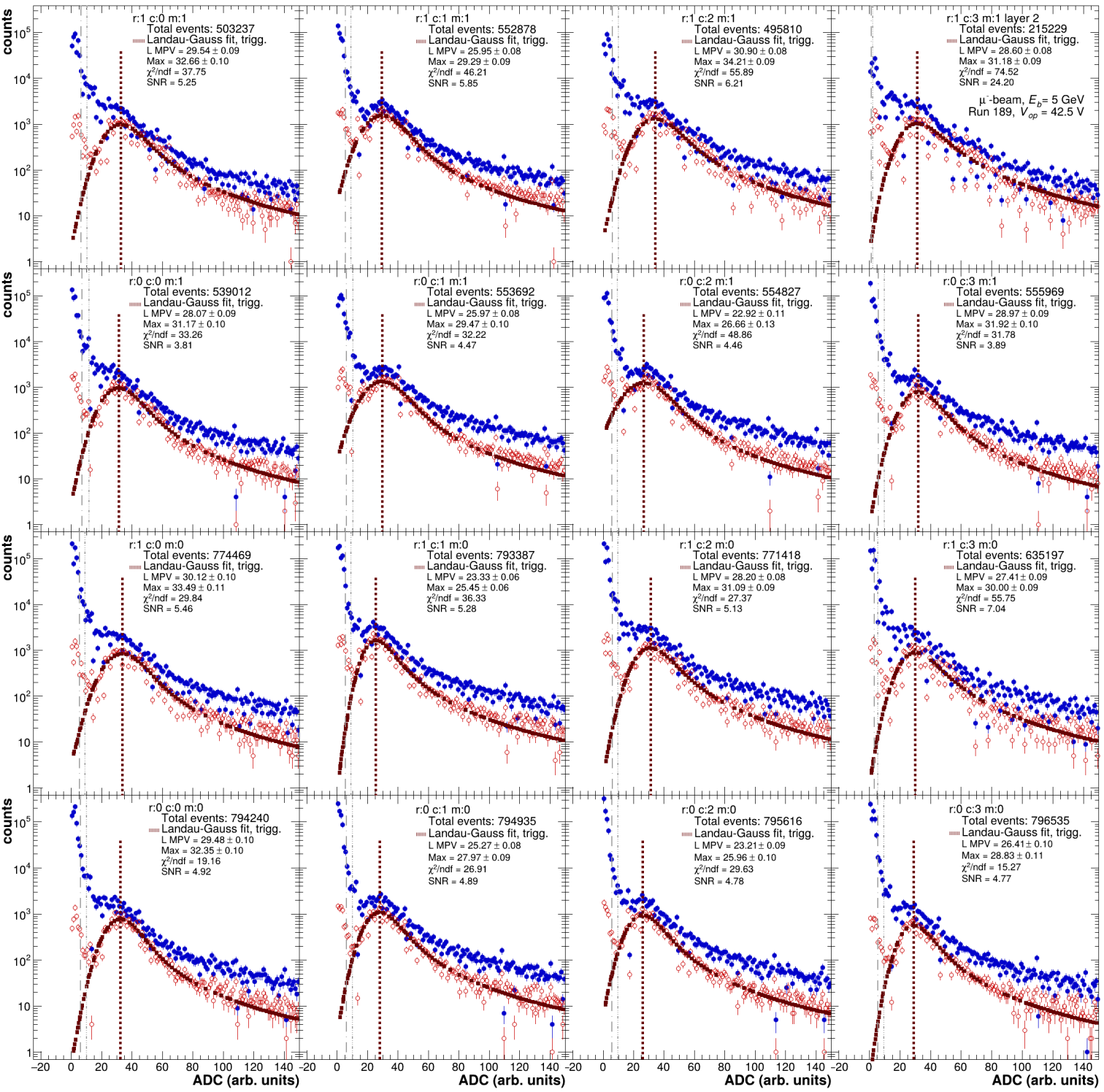

μ⁻-beam, E_b= 5 GeV, Run 189, V_{op} = 42.5 V
brd channel
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
1
1.5
2
2.5
3
3.5
4
4.5
5

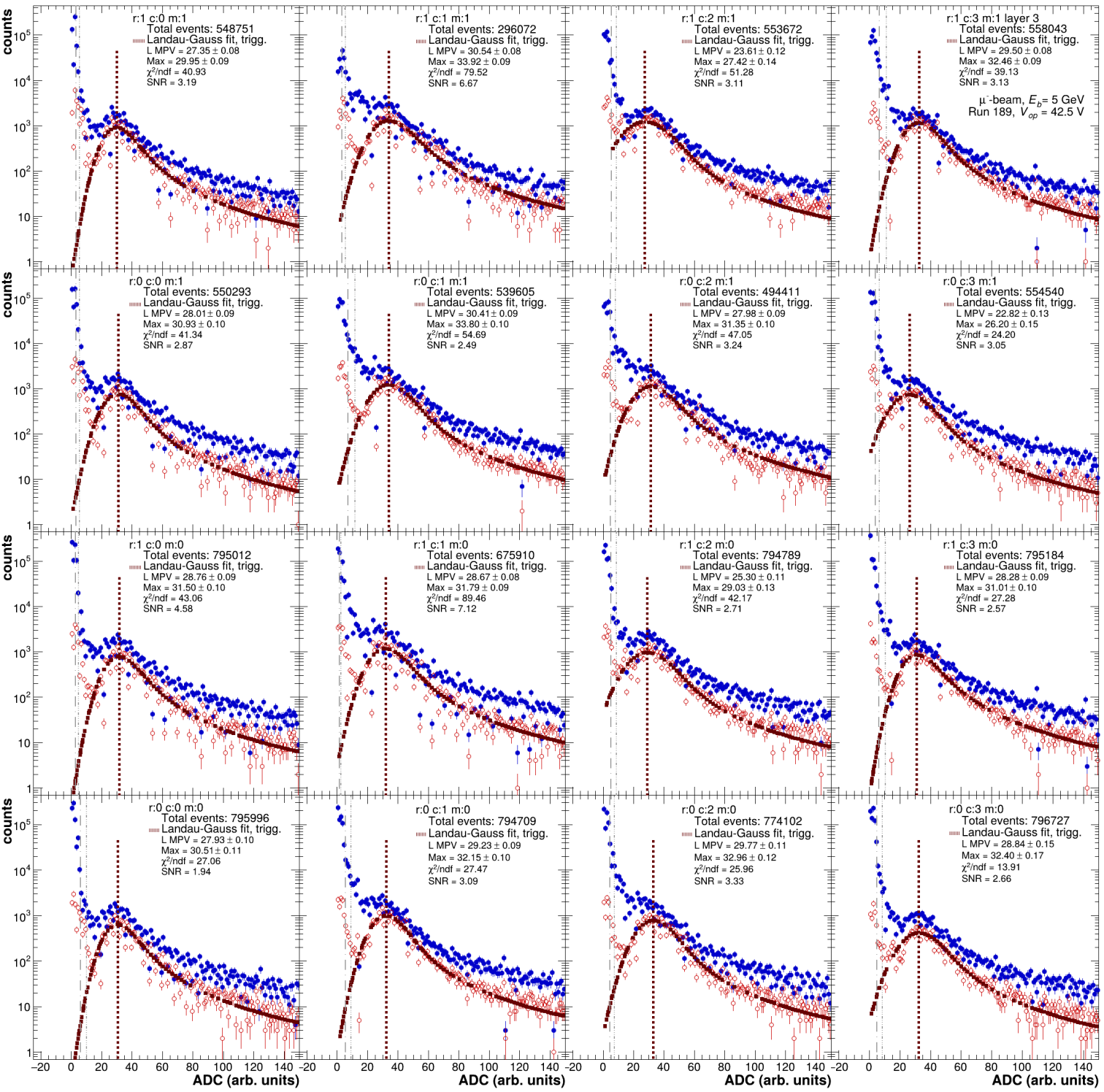

⟨ Max_HG ⟩ = 37.3
μ⁻-beam, E_b = 5 GeV, Run 189, V_op = 42.5 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
25
30
35
40
45
50
55
60

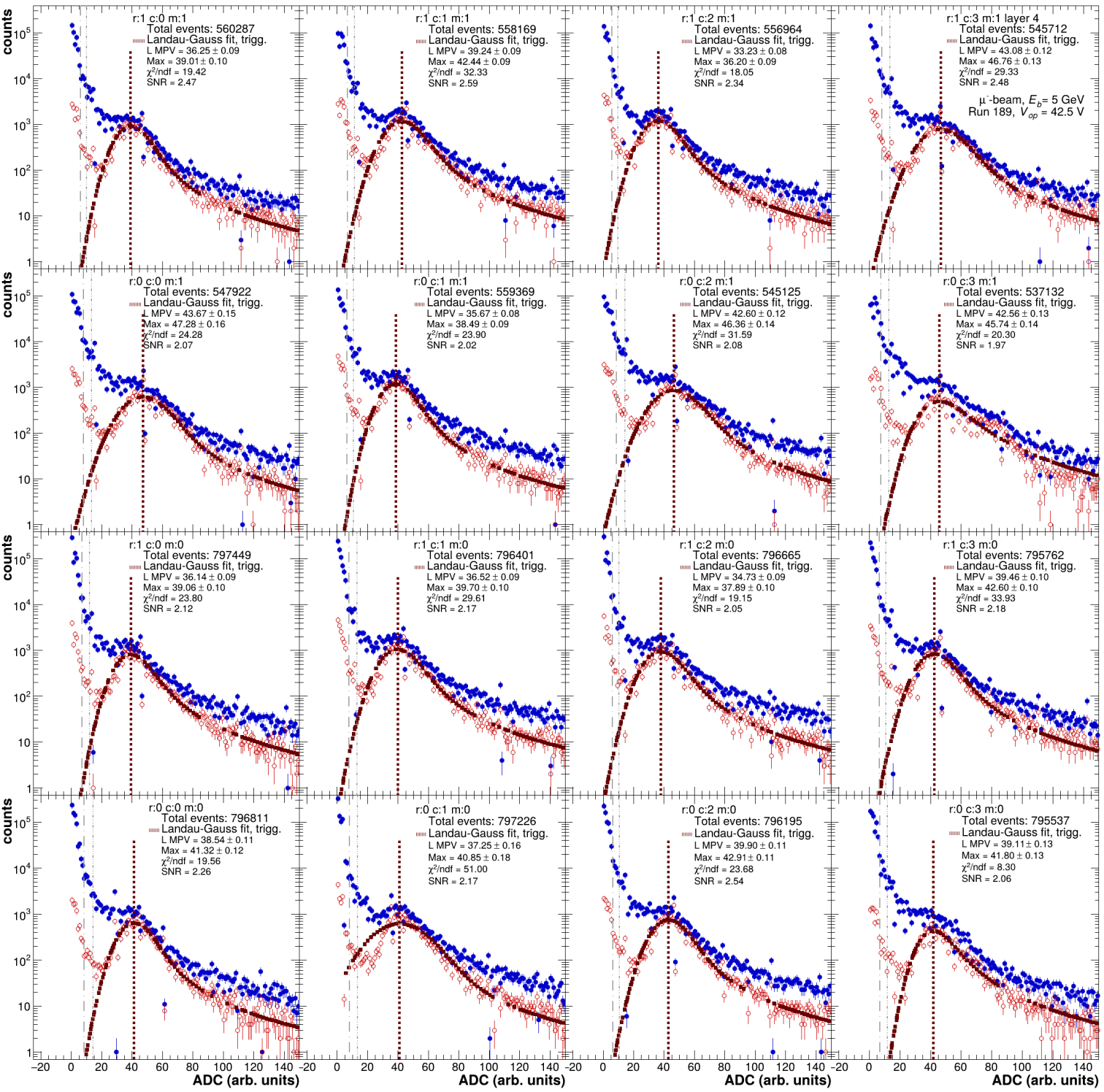

μ⁻-beam, E_b= 5 GeV, Run 189, V_{op} = 42.5 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50
60
70
80
90

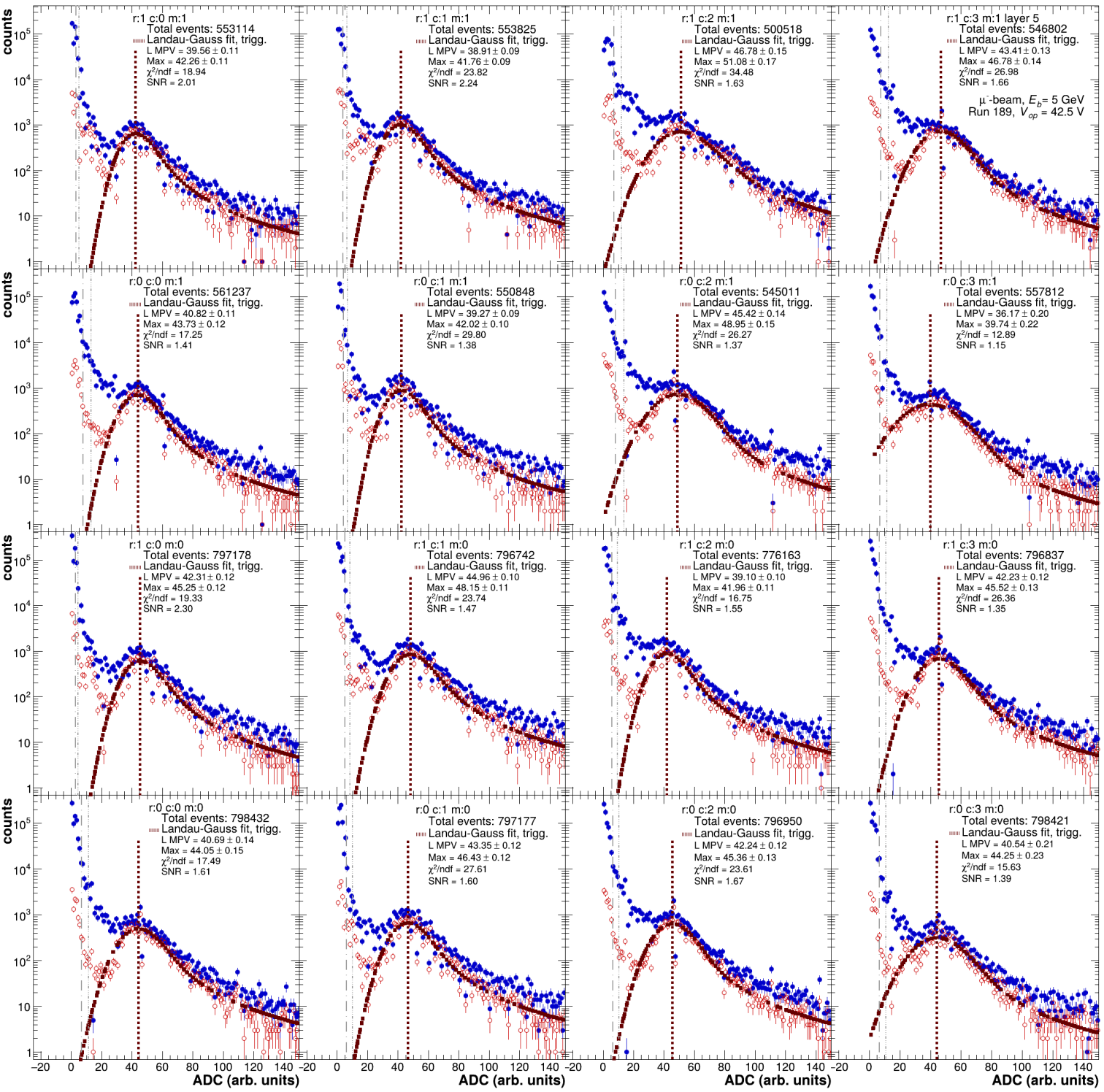

r:1 c:0 m:1
Total events: 539044
Landau-Gauss fit, trigg.
L MPV = 29.63 ± 0.08
Max = 32.29 ± 0.09
χ²/ndf = 31.23
SNR = 5.80
r:1 c:1 m:1
Total events: 500902
Landau-Gauss fit, trigg.
L MPV = 28.23 ± 0.06
Max = 30.66 ± 0.06
χ²/ndf = 58.86
SNR = 6.45
r:1 c:2 m:1
Total events: 453392
Landau-Gauss fit, trigg.
L MPV = 28.79 ± 0.06
Max = 31.35 ± 0.07
χ²/ndf = 71.59
SNR = 7.63
r:1 c:3 m:1 layer 0
Total events: 541905
Landau-Gauss fit, trigg.
L MPV = 29.89 ± 0.08
Max = 32.75 ± 0.09
χ²/ndf = 44.60
SNR = 5.83
μ⁻-beam, E_b = 5 GeV
Run 189, V_op = 42.5 V
r:0 c:0 m:1
Total events: 537739
Landau-Gauss fit, trigg.
L MPV = 29.12 ± 0.09
Max = 32.18 ± 0.10
χ²/ndf = 26.93
SNR = 3.34
r:0 c:1 m:1
Total events: 536313
Landau-Gauss fit, trigg.
L MPV = 26.00 ± 0.07
Max = 28.57 ± 0.07
χ²/ndf = 42.89
SNR = 3.07
r:0 c:2 m:1
Total events: 497610
Landau-Gauss fit, trigg.
L MPV = 27.61 ± 0.08
Max = 30.55 ± 0.08
χ²/ndf = 46.84
SNR = 3.12
r:0 c:3 m:1
Total events: 555907
Landau-Gauss fit, trigg.
L MPV = 23.55 ± 0.06
Max = 25.57 ± 0.07
χ²/ndf = 20.00
SNR = 2.48
r:1 c:0 m:0
Total events: 795325
Landau-Gauss fit, trigg.
L MPV = 16.79 ± 0.07
Max = 18.40 ± 0.08
χ²/ndf = 52.95
SNR = 4.13
r:1 c:1 m:0
Total events: 775147
Landau-Gauss fit, trigg.
L MPV = 18.87 ± 0.04
Max = 20.65 ± 0.05
χ²/ndf = 72.91
SNR = 4.63
r:1 c:2 m:0
r:1 c:3 m:0
Total events: 794912
Landau-Gauss fit, trigg.
L MPV = 16.24 ± 0.13
Max = 18.68 ± 0.14
χ²/ndf = 79.10
SNR = 4.24
r:0 c:0 m:0
Total events: 795207
Landau-Gauss fit, trigg.
L MPV = 18.61 ± 0.07
Max = 20.23 ± 0.07
χ²/ndf = 44.54
SNR = 4.95
r:0 c:1 m:0
Total events: 794757
Landau-Gauss fit, trigg.
L MPV = 18.58 ± 0.05
Max = 20.25 ± 0.05
χ²/ndf = 50.99
SNR = 4.78
r:0 c:2 m:0
Total events: 775528
Landau-Gauss fit, trigg.
L MPV = 18.55 ± 0.05
Max = 19.54 ± 0.05
χ²/ndf = 59.41
SNR = 5.67
r:0 c:3 m:0
Total events: 797788
Landau-Gauss fit, trigg.
L MPV = 16.19 ± 0.09
Max = 18.00 ± 0.10
χ²/ndf = 34.16
SNR = 3.18
counts
ADC (arb. units)

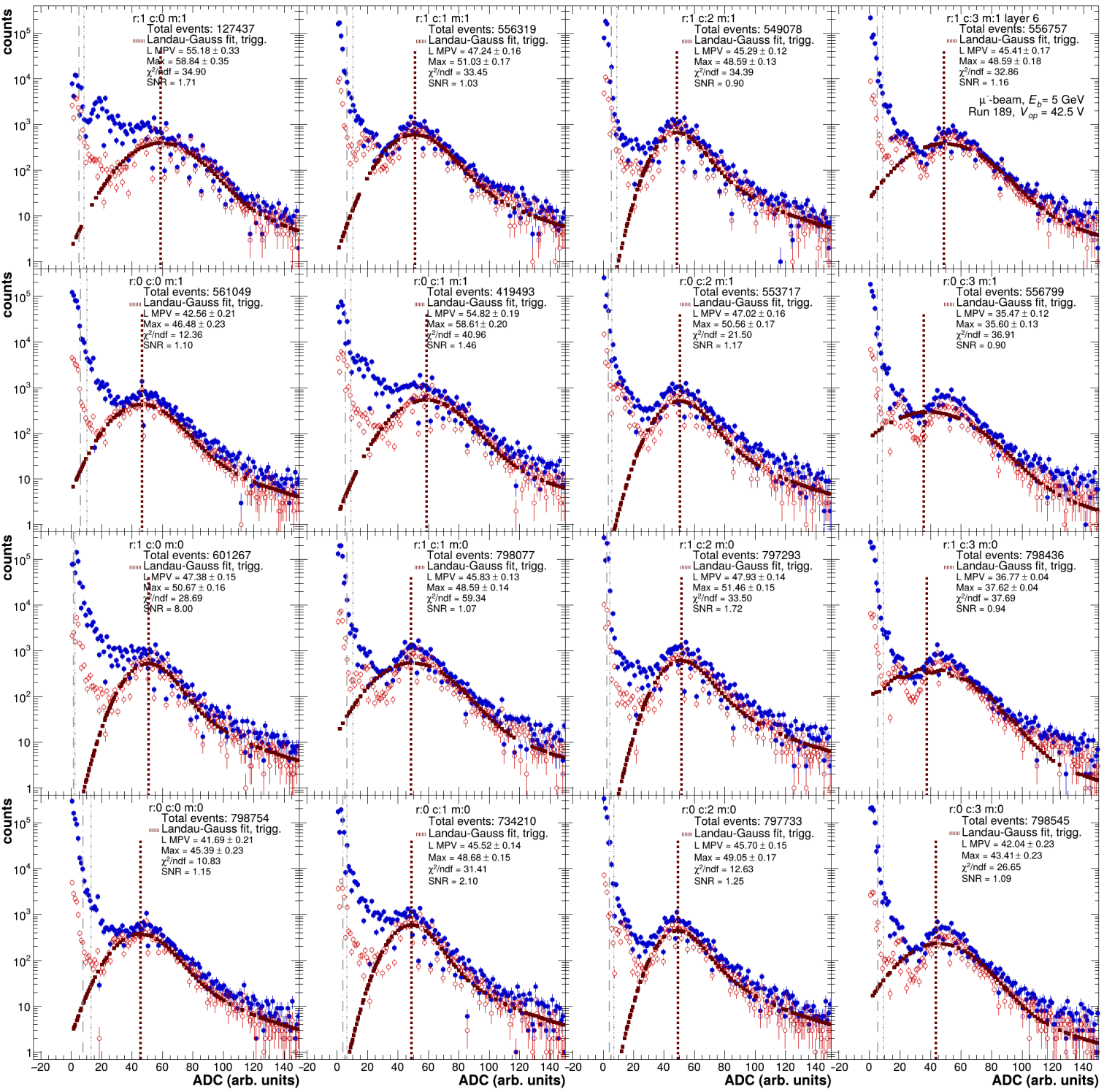

r:1 c:0 m:1
Total events: 536664
Landau-Gauss fit, trigg.
L MPV = 29.70 ± 0.09
Max = 32.59 ± 0.09
χ²/ndf = 32.86
SNR = 7.51
r:1 c:1 m:1
Total events: 536774
Landau-Gauss fit, trigg.
L MPV = 29.17 ± 0.07
Max = 31.97 ± 0.07
χ²/ndf = 40.37
SNR = 7.56
r:1 c:2 m:1
Total events: 538242
Landau-Gauss fit, trigg.
L MPV = 29.78 ± 0.07
Max = 32.73 ± 0.08
χ²/ndf = 52.04
SNR = 6.76
r:1 c:3 m:1 layer 1
Total events: 548445
Landau-Gauss fit, trigg.
L MPV = 25.31 ± 0.09
Max = 28.76 ± 0.11
χ²/ndf = 63.67
SNR = 5.92
μ⁻-beam, E_b = 5 GeV
Run 189, V_{op} = 42.5 V
r:0 c:0 m:1
Total events: 532892
Landau-Gauss fit, trigg.
L MPV = 28.84 ± 0.10
Max = 32.06 ± 0.11
χ²/ndf = 35.29
SNR = 5.57
r:0 c:1 m:1
Total events: 536856
Landau-Gauss fit, trigg.
L MPV = 29.87 ± 0.08
Max = 33.05 ± 0.08
χ²/ndf = 47.98
SNR = 5.13
r:0 c:2 m:1
Total events: 538629
Landau-Gauss fit, trigg.
L MPV = 28.96 ± 0.07
Max = 31.99 ± 0.08
χ²/ndf = 43.65
SNR = 4.78
r:0 c:3 m:1
r:1 c:0 m:0
Total events: 793847
Landau-Gauss fit, trigg.
L MPV = 19.21 ± 0.06
Max = 20.57 ± 0.07
χ²/ndf = 39.61
SNR = 5.94
r:1 c:1 m:0
Total events: 792379
Landau-Gauss fit, trigg.
L MPV = 20.01 ± 0.05
Max = 21.77 ± 0.06
χ²/ndf = 71.60
SNR = 6.50
r:1 c:2 m:0
Total events: 793403
Landau-Gauss fit, trigg.
L MPV = 18.26 ± 0.04
Max = 18.71 ± 0.04
χ²/ndf = 93.44
SNR = 5.49
r:1 c:3 m:0
Total events: 794797
Landau-Gauss fit, trigg.
L MPV = 16.49 ± 0.05
Max = 17.81 ± 0.06
χ²/ndf = 32.74
SNR = 5.08
r:0 c:0 m:0
Total events: 794301
Landau-Gauss fit, trigg.
L MPV = 18.99 ± 0.06
Max = 20.31 ± 0.07
χ²/ndf = 37.95
SNR = 6.45
r:0 c:1 m:0
Total events: 794110
Landau-Gauss fit, trigg.
L MPV = 21.97 ± 0.06
Max = 24.02 ± 0.07
χ²/ndf = 44.96
SNR = 6.59
r:0 c:2 m:0
Total events: 794958
Landau-Gauss fit, trigg.
L MPV = 20.01 ± 0.07
Max = 22.18 ± 0.08
χ²/ndf = 56.69
SNR = 7.22
r:0 c:3 m:0
Total events: 797231
Landau-Gauss fit, trigg.
L MPV = 18.42 ± 0.08
Max = 20.14 ± 0.08
χ²/ndf = 26.37
SNR = 4.31
counts
ADC (arb. units)

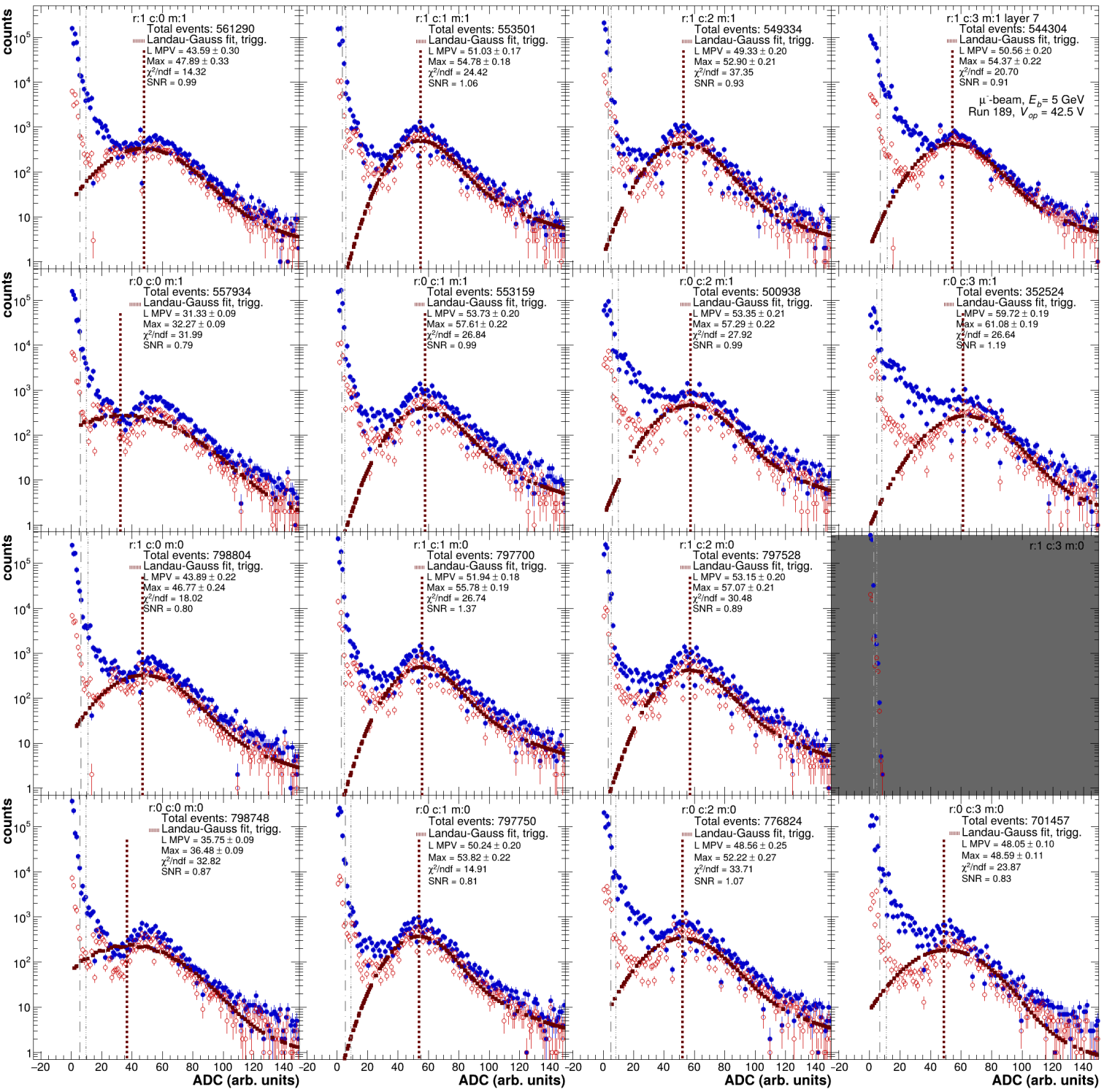

r:1 c:0 m:1
Total events: 503237
Landau-Gauss fit, trigg.
L MPV = 29.54 ± 0.09
Max = 32.66 ± 0.10
χ²/ndf = 37.75
SNR = 5.25
r:1 c:1 m:1
Total events: 552878
Landau-Gauss fit, trigg.
L MPV = 25.95 ± 0.08
Max = 29.29 ± 0.09
χ²/ndf = 46.21
SNR = 5.85
r:1 c:2 m:1
Total events: 495810
Landau-Gauss fit, trigg.
L MPV = 30.90 ± 0.08
Max = 34.21 ± 0.09
χ²/ndf = 55.89
SNR = 6.21
r:1 c:3 m:1 layer 2
Total events: 215229
Landau-Gauss fit, trigg.
L MPV = 28.60 ± 0.08
Max = 31.18 ± 0.09
χ²/ndf = 74.52
SNR = 24.20
μ⁻-beam, E_b = 5 GeV
Run 189, V_{op} = 42.5 V
r:0 c:0 m:1
Total events: 539012
Landau-Gauss fit, trigg.
L MPV = 28.07 ± 0.09
Max = 31.17 ± 0.10
χ²/ndf = 33.26
SNR = 3.81
r:0 c:1 m:1
Total events: 553692
Landau-Gauss fit, trigg.
L MPV = 25.97 ± 0.08
Max = 29.47 ± 0.10
χ²/ndf = 32.22
SNR = 4.47
r:0 c:2 m:1
Total events: 554827
Landau-Gauss fit, trigg.
L MPV = 22.92 ± 0.11
Max = 26.66 ± 0.13
χ²/ndf = 48.86
SNR = 4.46
r:0 c:3 m:1
Total events: 555969
Landau-Gauss fit, trigg.
L MPV = 28.97 ± 0.09
Max = 31.92 ± 0.10
χ²/ndf = 31.78
SNR = 3.89
r:1 c:0 m:0
Total events: 774469
Landau-Gauss fit, trigg.
L MPV = 30.12 ± 0.10
Max = 33.49 ± 0.11
χ²/ndf = 29.84
SNR = 5.46
r:1 c:1 m:0
Total events: 793387
Landau-Gauss fit, trigg.
L MPV = 23.33 ± 0.06
Max = 25.45 ± 0.06
χ²/ndf = 36.33
SNR = 5.28
r:1 c:2 m:0
Total events: 771418
Landau-Gauss fit, trigg.
L MPV = 28.20 ± 0.08
Max = 31.09 ± 0.09
χ²/ndf = 27.37
SNR = 5.13
r:1 c:3 m:0
Total events: 635197
Landau-Gauss fit, trigg.
L MPV = 27.41 ± 0.09
Max = 30.00 ± 0.09
χ²/ndf = 55.75
SNR = 7.04
r:0 c:0 m:0
Total events: 794240
Landau-Gauss fit, trigg.
L MPV = 29.48 ± 0.10
Max = 32.35 ± 0.10
χ²/ndf = 19.16
SNR = 4.92
r:0 c:1 m:0
Total events: 794935
Landau-Gauss fit, trigg.
L MPV = 25.27 ± 0.08
Max = 27.97 ± 0.09
χ²/ndf = 26.91
SNR = 4.89
r:0 c:2 m:0
Total events: 795616
Landau-Gauss fit, trigg.
L MPV = 23.21 ± 0.09
Max = 25.96 ± 0.10
χ²/ndf = 29.63
SNR = 4.78
r:0 c:3 m:0
Total events: 796535
Landau-Gauss fit, trigg.
L MPV = 26.41 ± 0.10
Max = 28.83 ± 0.11
χ²/ndf = 15.27
SNR = 4.77
counts
ADC (arb. units)

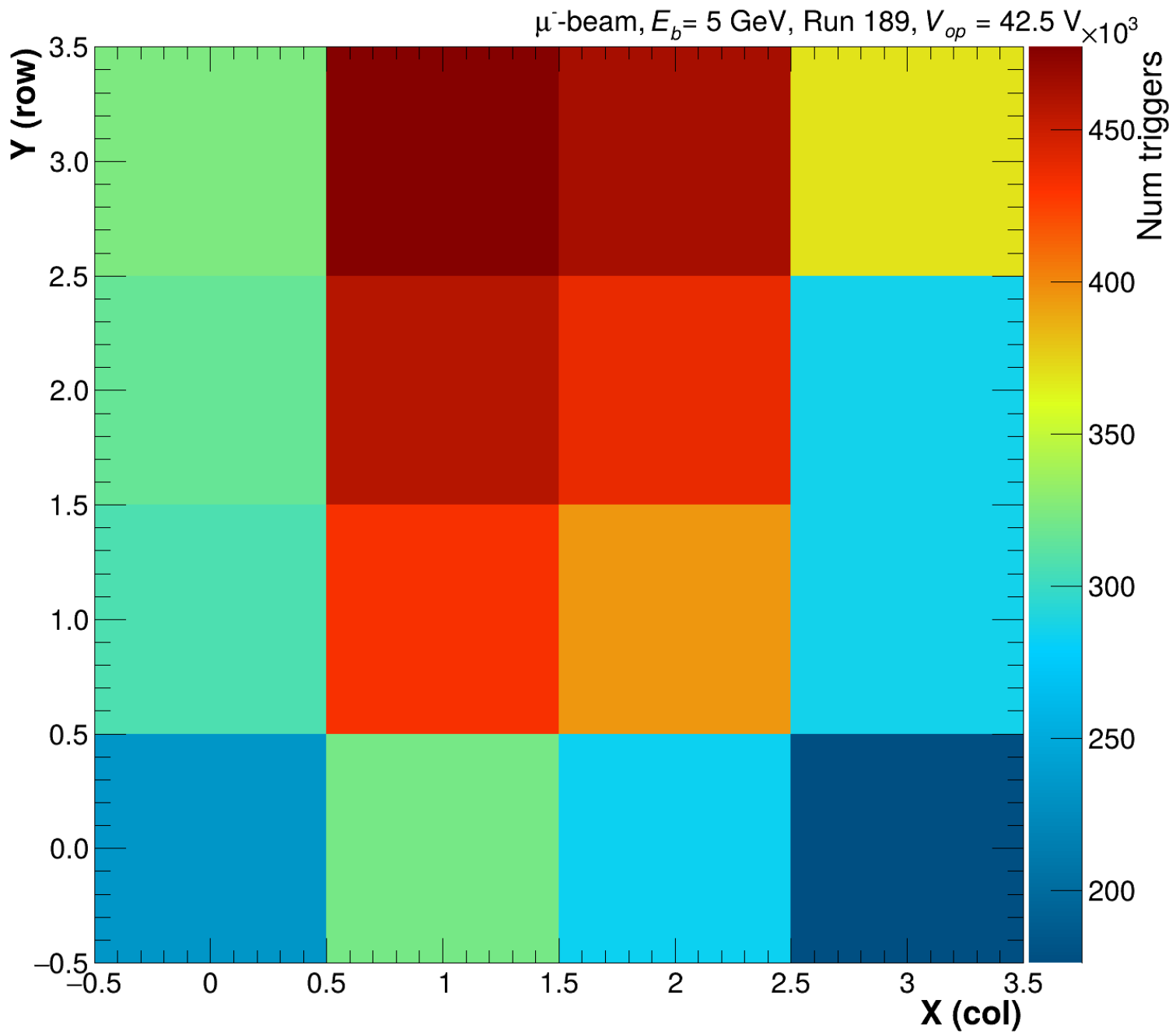

r:1 c:0 m:1
Total events: 548751
Landau-Gauss fit, trigg.
L MPV = 27.35 ± 0.08
Max = 29.95 ± 0.09
χ²/ndf = 40.93
SNR = 3.19
r:1 c:1 m:1
Total events: 296072
Landau-Gauss fit, trigg.
L MPV = 30.54 ± 0.08
Max = 33.92 ± 0.09
χ²/ndf = 79.52
SNR = 6.67
r:1 c:2 m:1
Total events: 553672
Landau-Gauss fit, trigg.
L MPV = 23.61 ± 0.12
Max = 27.42 ± 0.14
χ²/ndf = 51.28
SNR = 3.11
r:1 c:3 m:1 layer 3
Total events: 558043
Landau-Gauss fit, trigg.
L MPV = 29.50 ± 0.08
Max = 32.46 ± 0.09
χ²/ndf = 39.13
SNR = 3.13
μ⁻-beam, E_b = 5 GeV
Run 189, V_op = 42.5 V
r:0 c:0 m:1
Total events: 550293
Landau-Gauss fit, trigg.
L MPV = 28.01 ± 0.09
Max = 30.93 ± 0.10
χ²/ndf = 41.34
SNR = 2.87
r:0 c:1 m:1
Total events: 539605
Landau-Gauss fit, trigg.
L MPV = 30.41 ± 0.09
Max = 33.80 ± 0.10
χ²/ndf = 54.69
SNR = 2.49
r:0 c:2 m:1
Total events: 494411
Landau-Gauss fit, trigg.
L MPV = 27.98 ± 0.09
Max = 31.35 ± 0.10
χ²/ndf = 47.05
SNR = 3.24
r:0 c:3 m:1
Total events: 554540
Landau-Gauss fit, trigg.
L MPV = 22.82 ± 0.13
Max = 26.20 ± 0.15
χ²/ndf = 24.20
SNR = 3.05
r:1 c:0 m:0
Total events: 795012
Landau-Gauss fit, trigg.
L MPV = 28.76 ± 0.09
Max = 31.50 ± 0.10
χ²/ndf = 43.06
SNR = 4.58
r:1 c:1 m:0
Total events: 675910
Landau-Gauss fit, trigg.
L MPV = 28.67 ± 0.08
Max = 31.79 ± 0.09
χ²/ndf = 89.46
SNR = 7.12
r:1 c:2 m:0
Total events: 794789
Landau-Gauss fit, trigg.
L MPV = 25.30 ± 0.11
Max = 29.03 ± 0.13
χ²/ndf = 42.17
SNR = 2.71
r:1 c:3 m:0
Total events: 795184
Landau-Gauss fit, trigg.
L MPV = 28.28 ± 0.09
Max = 31.01 ± 0.10
χ²/ndf = 27.28
SNR = 2.57
r:0 c:0 m:0
Total events: 795996
Landau-Gauss fit, trigg.
L MPV = 27.93 ± 0.10
Max = 30.51 ± 0.11
χ²/ndf = 27.06
SNR = 1.94
r:0 c:1 m:0
Total events: 794709
Landau-Gauss fit, trigg.
L MPV = 29.23 ± 0.09
Max = 32.15 ± 0.10
χ²/ndf = 27.47
SNR = 3.09
r:0 c:2 m:0
Total events: 774102
Landau-Gauss fit, trigg.
L MPV = 29.77 ± 0.11
Max = 32.96 ± 0.12
χ²/ndf = 25.96
SNR = 3.33
r:0 c:3 m:0
Total events: 796727
Landau-Gauss fit, trigg.
L MPV = 28.84 ± 0.15
Max = 32.40 ± 0.17
χ²/ndf = 13.91
SNR = 2.66
counts
ADC (arb. units)

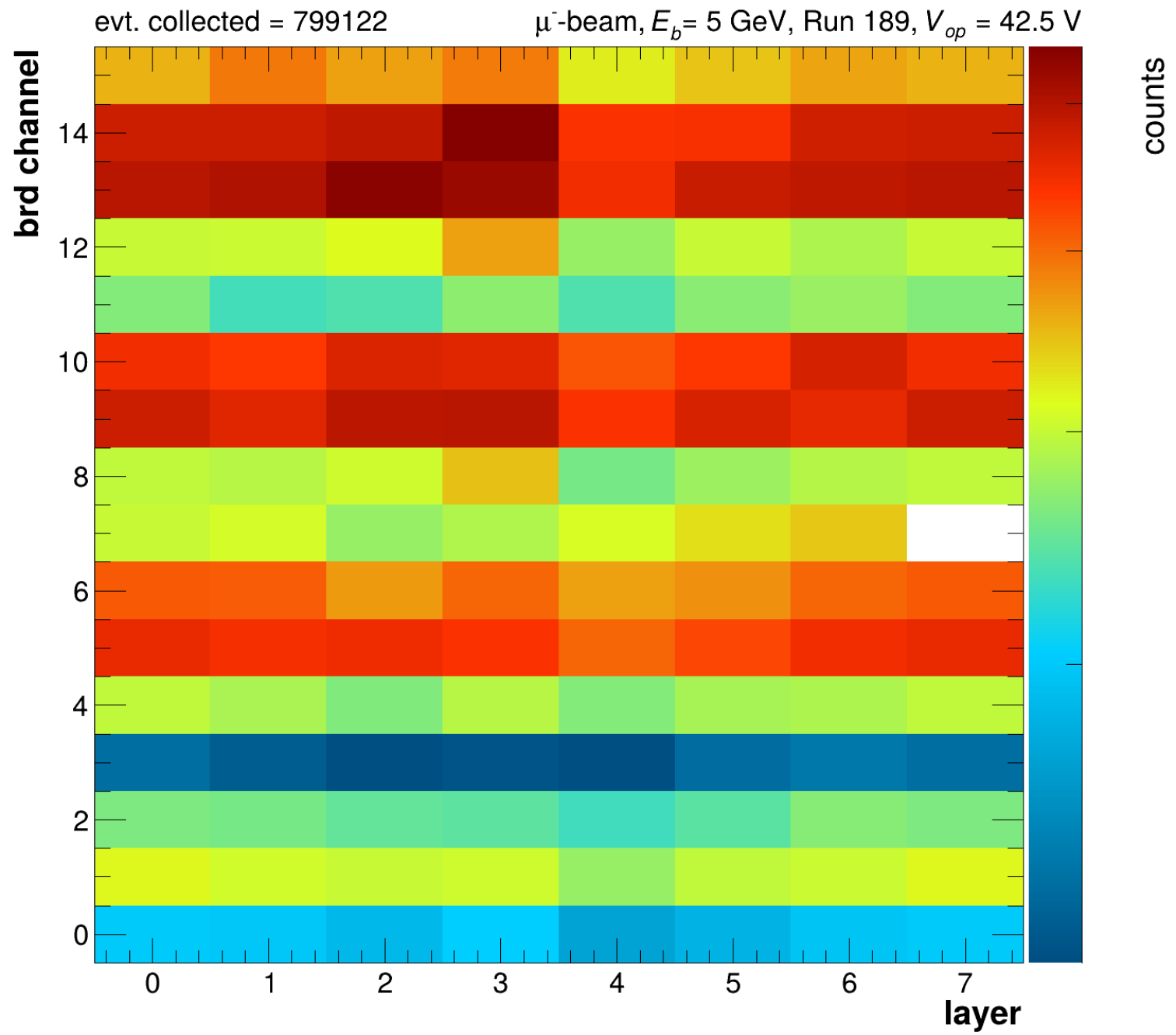

counts
ADC (arb. units)
r:1 c:0 m:1
Total events: 560287
Landau-Gauss fit, trigg.
L MPV = 36.25 ± 0.09
Max = 39.01 ± 0.10
χ²/ndf = 19.42
SNR = 2.47
r:1 c:1 m:1
Total events: 558169
Landau-Gauss fit, trigg.
L MPV = 39.24 ± 0.09
Max = 42.44 ± 0.09
χ²/ndf = 32.33
SNR = 2.59
r:1 c:2 m:1
Total events: 556964
Landau-Gauss fit, trigg.
L MPV = 33.23 ± 0.08
Max = 36.20 ± 0.09
χ²/ndf = 18.05
SNR = 2.34
r:1 c:3 m:1 layer 4
Total events: 545712
Landau-Gauss fit, trigg.
L MPV = 43.08 ± 0.12
Max = 46.76 ± 0.13
χ²/ndf = 29.33
SNR = 2.48
μ-beam, E_b = 5 GeV
Run 189, V_op = 42.5 V
r:0 c:0 m:1
Total events: 547922
Landau-Gauss fit, trigg.
L MPV = 43.67 ± 0.15
Max = 47.28 ± 0.16
χ²/ndf = 24.28
SNR = 2.07
r:0 c:1 m:1
Total events: 559369
Landau-Gauss fit, trigg.
L MPV = 35.67 ± 0.08
Max = 38.49 ± 0.09
χ²/ndf = 23.90
SNR = 2.02
r:0 c:2 m:1
Total events: 545125
Landau-Gauss fit, trigg.
L MPV = 42.60 ± 0.12
Max = 46.36 ± 0.14
χ²/ndf = 31.59
SNR = 2.08
r:0 c:3 m:1
Total events: 537132
Landau-Gauss fit, trigg.
L MPV = 42.56 ± 0.13
Max = 45.74 ± 0.14
χ²/ndf = 20.30
SNR = 1.97
r:1 c:0 m:0
Total events: 797449
Landau-Gauss fit, trigg.
L MPV = 36.14 ± 0.09
Max = 39.06 ± 0.10
χ²/ndf = 23.80
SNR = 2.12
r:1 c:1 m:0
Total events: 796401
Landau-Gauss fit, trigg.
L MPV = 36.52 ± 0.09
Max = 39.70 ± 0.10
χ²/ndf = 29.61
SNR = 2.17
r:1 c:2 m:0
Total events: 796665
Landau-Gauss fit, trigg.
L MPV = 34.73 ± 0.09
Max = 37.89 ± 0.10
χ²/ndf = 19.15
SNR = 2.05
r:1 c:3 m:0
Total events: 795762
Landau-Gauss fit, trigg.
L MPV = 39.46 ± 0.10
Max = 42.60 ± 0.10
χ²/ndf = 33.93
SNR = 2.18
r:0 c:0 m:0
Total events: 796811
Landau-Gauss fit, trigg.
L MPV = 38.54 ± 0.11
Max = 41.32 ± 0.12
χ²/ndf = 19.56
SNR = 2.26
r:0 c:1 m:0
Total events: 797226
Landau-Gauss fit, trigg.
L MPV = 37.25 ± 0.16
Max = 40.85 ± 0.18
χ²/ndf = 51.00
SNR = 2.17
r:0 c:2 m:0
Total events: 796195
Landau-Gauss fit, trigg.
L MPV = 39.90 ± 0.11
Max = 42.91 ± 0.11
χ²/ndf = 23.68
SNR = 2.54
r:0 c:3 m:0
Total events: 795537
Landau-Gauss fit, trigg.
L MPV = 39.11 ± 0.13
Max = 41.80 ± 0.13
χ²/ndf = 8.30
SNR = 2.06

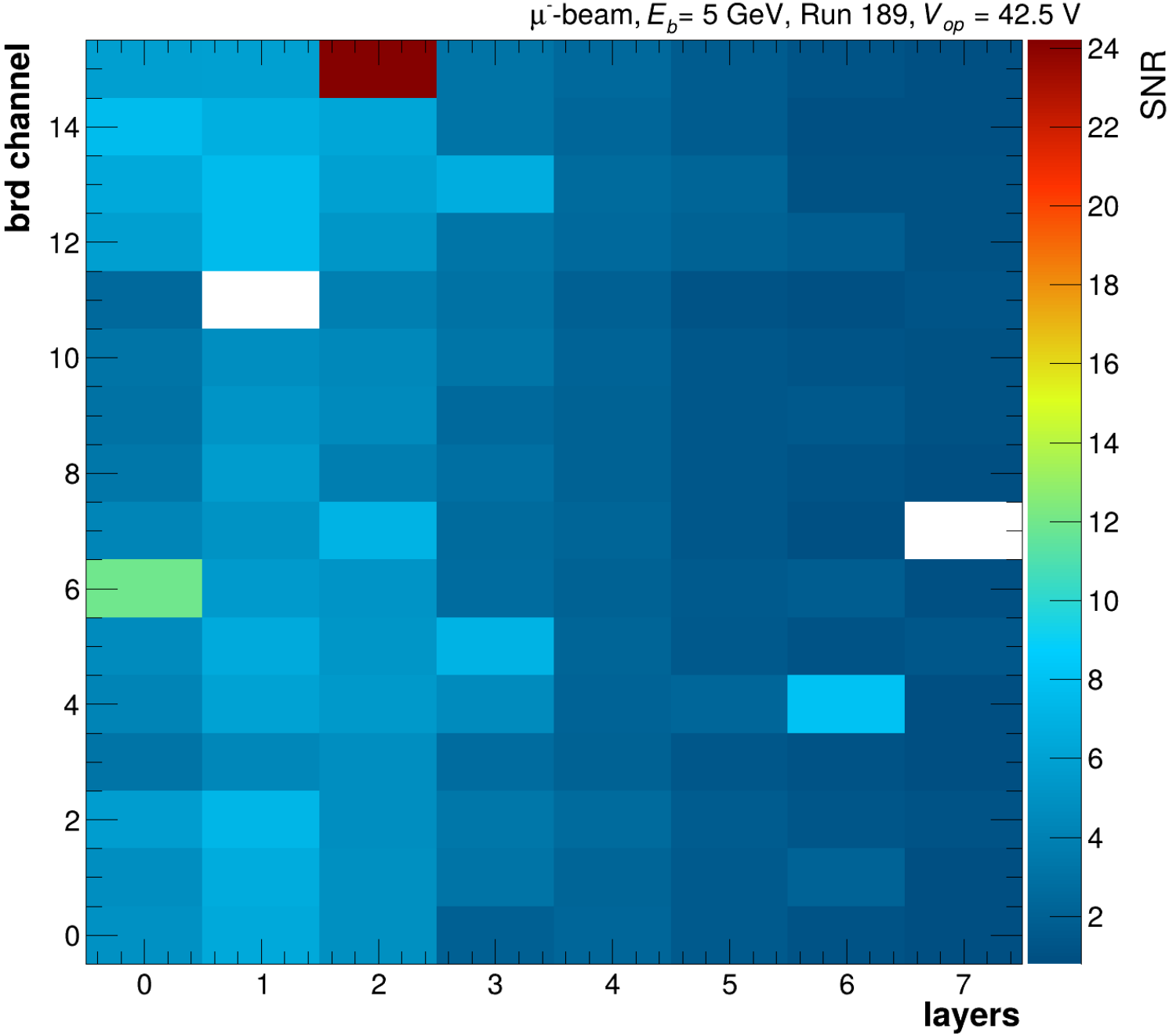

r:1 c:0 m:1
Total events: 553114
Landau-Gauss fit, trigg.
L MPV = 39.56 ± 0.11
Max = 42.26 ± 0.11
χ²/ndf = 18.94
SNR = 2.01
r:1 c:1 m:1
Total events: 553825
Landau-Gauss fit, trigg.
L MPV = 38.91 ± 0.09
Max = 41.76 ± 0.09
χ²/ndf = 23.82
SNR = 2.24
r:1 c:2 m:1
Total events: 500518
Landau-Gauss fit, trigg.
L MPV = 46.78 ± 0.15
Max = 51.08 ± 0.17
χ²/ndf = 34.48
SNR = 1.63
r:1 c:3 m:1 layer 5
Total events: 546802
Landau-Gauss fit, trigg.
L MPV = 43.41 ± 0.13
Max = 46.78 ± 0.14
χ²/ndf = 26.98
SNR = 1.66
μ⁻-beam, E_b = 5 GeV
Run 189, V_op = 42.5 V
r:0 c:0 m:1
Total events: 561237
Landau-Gauss fit, trigg.
L MPV = 40.82 ± 0.11
Max = 43.73 ± 0.12
χ²/ndf = 17.25
SNR = 1.41
r:0 c:1 m:1
Total events: 550848
Landau-Gauss fit, trigg.
L MPV = 39.27 ± 0.09
Max = 42.02 ± 0.10
χ²/ndf = 29.80
SNR = 1.38
r:0 c:2 m:1
Total events: 545011
Landau-Gauss fit, trigg.
L MPV = 45.42 ± 0.14
Max = 48.95 ± 0.15
χ²/ndf = 26.27
SNR = 1.37
r:0 c:3 m:1
Total events: 557812
Landau-Gauss fit, trigg.
L MPV = 36.17 ± 0.20
Max = 39.74 ± 0.22
χ²/ndf = 12.89
SNR = 1.15
r:1 c:0 m:0
Total events: 797178
Landau-Gauss fit, trigg.
L MPV = 42.31 ± 0.12
Max = 45.25 ± 0.12
χ²/ndf = 19.33
SNR = 2.30
r:1 c:1 m:0
Total events: 796742
Landau-Gauss fit, trigg.
L MPV = 44.96 ± 0.10
Max = 48.15 ± 0.11
χ²/ndf = 23.74
SNR = 1.47
r:1 c:2 m:0
Total events: 776163
Landau-Gauss fit, trigg.
L MPV = 39.10 ± 0.10
Max = 41.96 ± 0.11
χ²/ndf = 16.75
SNR = 1.55
r:1 c:3 m:0
Total events: 796837
Landau-Gauss fit, trigg.
L MPV = 42.23 ± 0.12
Max = 45.52 ± 0.13
χ²/ndf = 26.36
SNR = 1.35
r:0 c:0 m:0
Total events: 798432
Landau-Gauss fit, trigg.
L MPV = 40.69 ± 0.14
Max = 44.05 ± 0.15
χ²/ndf = 17.49
SNR = 1.61
r:0 c:1 m:0
Total events: 797177
Landau-Gauss fit, trigg.
L MPV = 43.35 ± 0.12
Max = 46.43 ± 0.12
χ²/ndf = 27.61
SNR = 1.60
r:0 c:2 m:0
Total events: 796950
Landau-Gauss fit, trigg.
L MPV = 42.24 ± 0.12
Max = 45.36 ± 0.13
χ²/ndf = 23.61
SNR = 1.67
r:0 c:3 m:0
Total events: 798421
Landau-Gauss fit, trigg.
L MPV = 40.54 ± 0.21
Max = 44.25 ± 0.23
χ²/ndf = 15.63
SNR = 1.39
counts
ADC (arb. units)

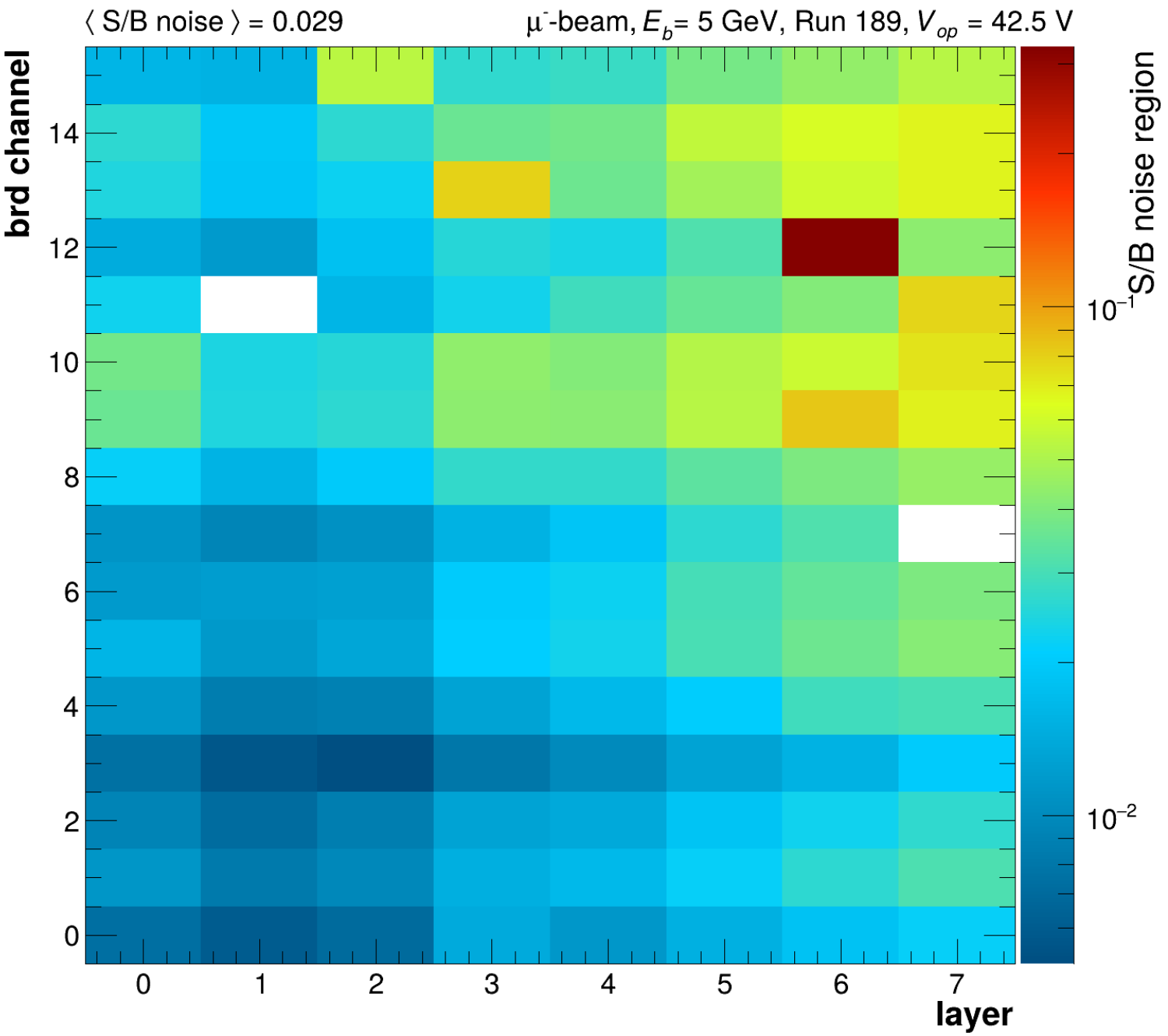

r:1 c:0 m:1
Total events: 127437
Landau-Gauss fit, trigg.
L MPV = 55.18 ± 0.33
Max = 58.84 ± 0.35
χ²/ndf = 34.90
SNR = 1.71
r:1 c:1 m:1
Total events: 556319
Landau-Gauss fit, trigg.
L MPV = 47.24 ± 0.16
Max = 51.03 ± 0.17
χ²/ndf = 33.45
SNR = 1.03
r:1 c:2 m:1
Total events: 549078
Landau-Gauss fit, trigg.
L MPV = 45.29 ± 0.12
Max = 48.59 ± 0.13
χ²/ndf = 34.39
SNR = 0.90
r:1 c:3 m:1 layer 6
Total events: 556757
Landau-Gauss fit, trigg.
L MPV = 45.41 ± 0.17
Max = 48.59 ± 0.18
χ²/ndf = 32.86
SNR = 1.16
μ-beam, E_b = 5 GeV
Run 189, V_op = 42.5 V
r:0 c:0 m:1
Total events: 561049
Landau-Gauss fit, trigg.
L MPV = 42.56 ± 0.21
Max = 46.48 ± 0.23
χ²/ndf = 12.36
SNR = 1.10
r:0 c:1 m:1
Total events: 419493
Landau-Gauss fit, trigg.
L MPV = 54.82 ± 0.19
Max = 58.61 ± 0.20
χ²/ndf = 40.96
SNR = 1.46
r:0 c:2 m:1
Total events: 553717
Landau-Gauss fit, trigg.
L MPV = 47.02 ± 0.16
Max = 50.56 ± 0.17
χ²/ndf = 21.50
SNR = 1.17
r:0 c:3 m:1
Total events: 556799
Landau-Gauss fit, trigg.
L MPV = 35.47 ± 0.12
Max = 35.60 ± 0.13
χ²/ndf = 36.91
SNR = 0.90
r:1 c:0 m:0
Total events: 601267
Landau-Gauss fit, trigg.
L MPV = 47.38 ± 0.15
Max = 50.67 ± 0.16
χ²/ndf = 28.69
SNR = 8.00
r:1 c:1 m:0
Total events: 798077
Landau-Gauss fit, trigg.
L MPV = 45.83 ± 0.13
Max = 48.59 ± 0.14
χ²/ndf = 59.34
SNR = 1.07
r:1 c:2 m:0
Total events: 797293
Landau-Gauss fit, trigg.
L MPV = 47.93 ± 0.14
Max = 51.46 ± 0.15
χ²/ndf = 33.50
SNR = 1.72
r:1 c:3 m:0
Total events: 798436
Landau-Gauss fit, trigg.
L MPV = 36.77 ± 0.04
Max = 37.62 ± 0.04
χ²/ndf = 37.69
SNR = 0.94
r:0 c:0 m:0
Total events: 798754
Landau-Gauss fit, trigg.
L MPV = 41.69 ± 0.21
Max = 45.39 ± 0.23
χ²/ndf = 10.83
SNR = 1.15
r:0 c:1 m:0
Total events: 734210
Landau-Gauss fit, trigg.
L MPV = 45.52 ± 0.14
Max = 48.68 ± 0.15
χ²/ndf = 31.41
SNR = 2.10
r:0 c:2 m:0
Total events: 797733
Landau-Gauss fit, trigg.
L MPV = 45.70 ± 0.15
Max = 49.05 ± 0.17
χ²/ndf = 12.63
SNR = 1.25
r:0 c:3 m:0
Total events: 798545
Landau-Gauss fit, trigg.
L MPV = 42.04 ± 0.23
Max = 43.41 ± 0.23
χ²/ndf = 26.65
SNR = 1.09
counts
ADC (arb. units)

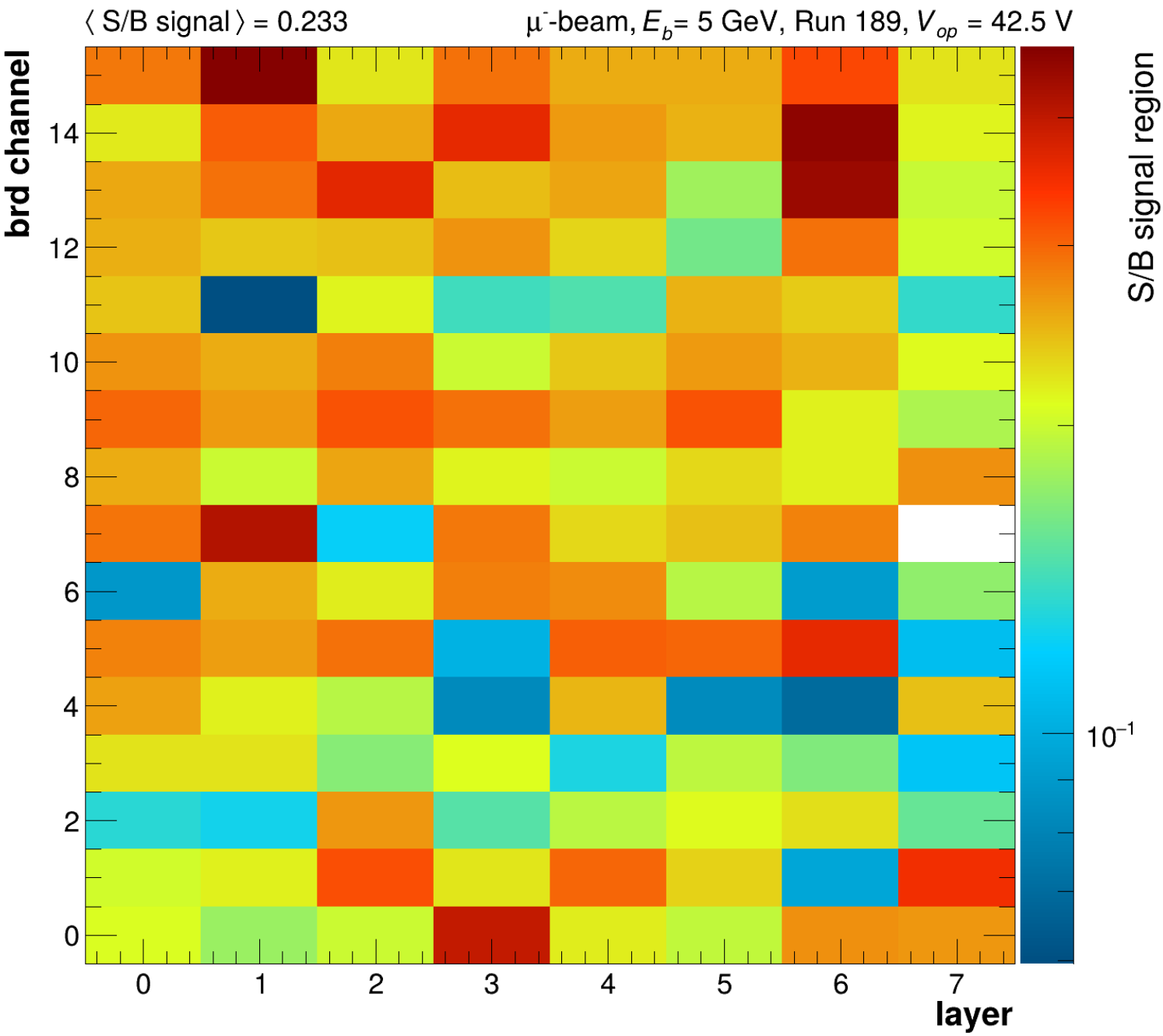

r:1 c:0 m:1
Total events: 561290
Landau-Gauss fit, trigg.
L MPV = 43.59 ± 0.30
Max = 47.89 ± 0.33
χ²/ndf = 14.32
SNR = 0.99
r:1 c:1 m:1
Total events: 553501
Landau-Gauss fit, trigg.
L MPV = 51.03 ± 0.17
Max = 54.78 ± 0.18
χ²/ndf = 24.42
SNR = 1.06
r:1 c:2 m:1
Total events: 549334
Landau-Gauss fit, trigg.
L MPV = 49.33 ± 0.20
Max = 52.90 ± 0.21
χ²/ndf = 37.35
SNR = 0.93
r:1 c:3 m:1 layer 7
Total events: 544304
Landau-Gauss fit, trigg.
L MPV = 50.56 ± 0.20
Max = 54.37 ± 0.22
χ²/ndf = 20.70
SNR = 0.91
μ⁻-beam, E_b = 5 GeV
Run 189, V_{op} = 42.5 V
r:0 c:0 m:1
Total events: 557934
Landau-Gauss fit, trigg.
L MPV = 31.33 ± 0.09
Max = 32.27 ± 0.09
χ²/ndf = 31.99
SNR = 0.79
r:0 c:1 m:1
Total events: 553159
Landau-Gauss fit, trigg.
L MPV = 53.73 ± 0.20
Max = 57.61 ± 0.22
χ²/ndf = 26.84
SNR = 0.99
r:0 c:2 m:1
Total events: 500938
Landau-Gauss fit, trigg.
L MPV = 53.35 ± 0.21
Max = 57.29 ± 0.22
χ²/ndf = 27.92
SNR = 0.99
r:0 c:3 m:1
Total events: 352524
Landau-Gauss fit, trigg.
L MPV = 59.72 ± 0.19
Max = 61.08 ± 0.19
χ²/ndf = 26.64
SNR = 1.19
r:1 c:0 m:0
Total events: 798804
Landau-Gauss fit, trigg.
L MPV = 43.89 ± 0.22
Max = 46.77 ± 0.24
χ²/ndf = 18.02
SNR = 0.80
r:1 c:1 m:0
Total events: 797700
Landau-Gauss fit, trigg.
L MPV = 51.94 ± 0.18
Max = 55.78 ± 0.19
χ²/ndf = 26.74
SNR = 1.37
r:1 c:2 m:0
Total events: 797528
Landau-Gauss fit, trigg.
L MPV = 53.15 ± 0.20
Max = 57.07 ± 0.21
χ²/ndf = 30.48
SNR = 0.89
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 798748
Landau-Gauss fit, trigg.
L MPV = 35.75 ± 0.09
Max = 36.48 ± 0.09
χ²/ndf = 32.82
SNR = 0.87
r:0 c:1 m:0
Total events: 797750
Landau-Gauss fit, trigg.
L MPV = 50.24 ± 0.20
Max = 53.82 ± 0.22
χ²/ndf = 14.91
SNR = 0.81
r:0 c:2 m:0
Total events: 776824
Landau-Gauss fit, trigg.
L MPV = 48.56 ± 0.25
Max = 52.22 ± 0.27
χ²/ndf = 33.71
SNR = 1.07
r:0 c:3 m:0
Total events: 701457
Landau-Gauss fit, trigg.
L MPV = 48.05 ± 0.10
Max = 48.59 ± 0.11
χ²/ndf = 23.87
SNR = 0.83
counts
ADC (arb. units)

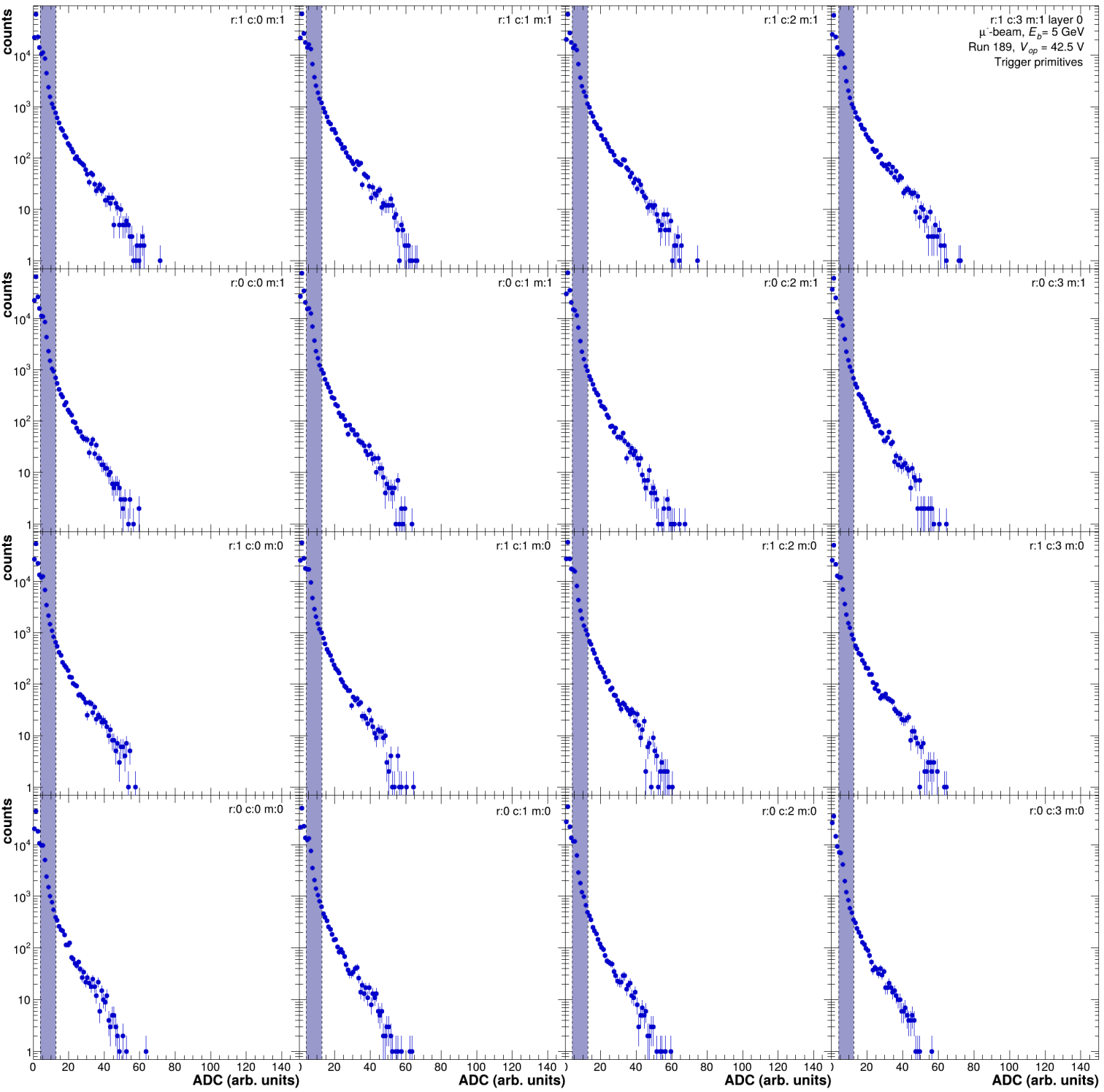

μ⁻-beam, E_b= 5 GeV, Run 189, V_{op} = 42.5 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
450
400
350
300
250
200

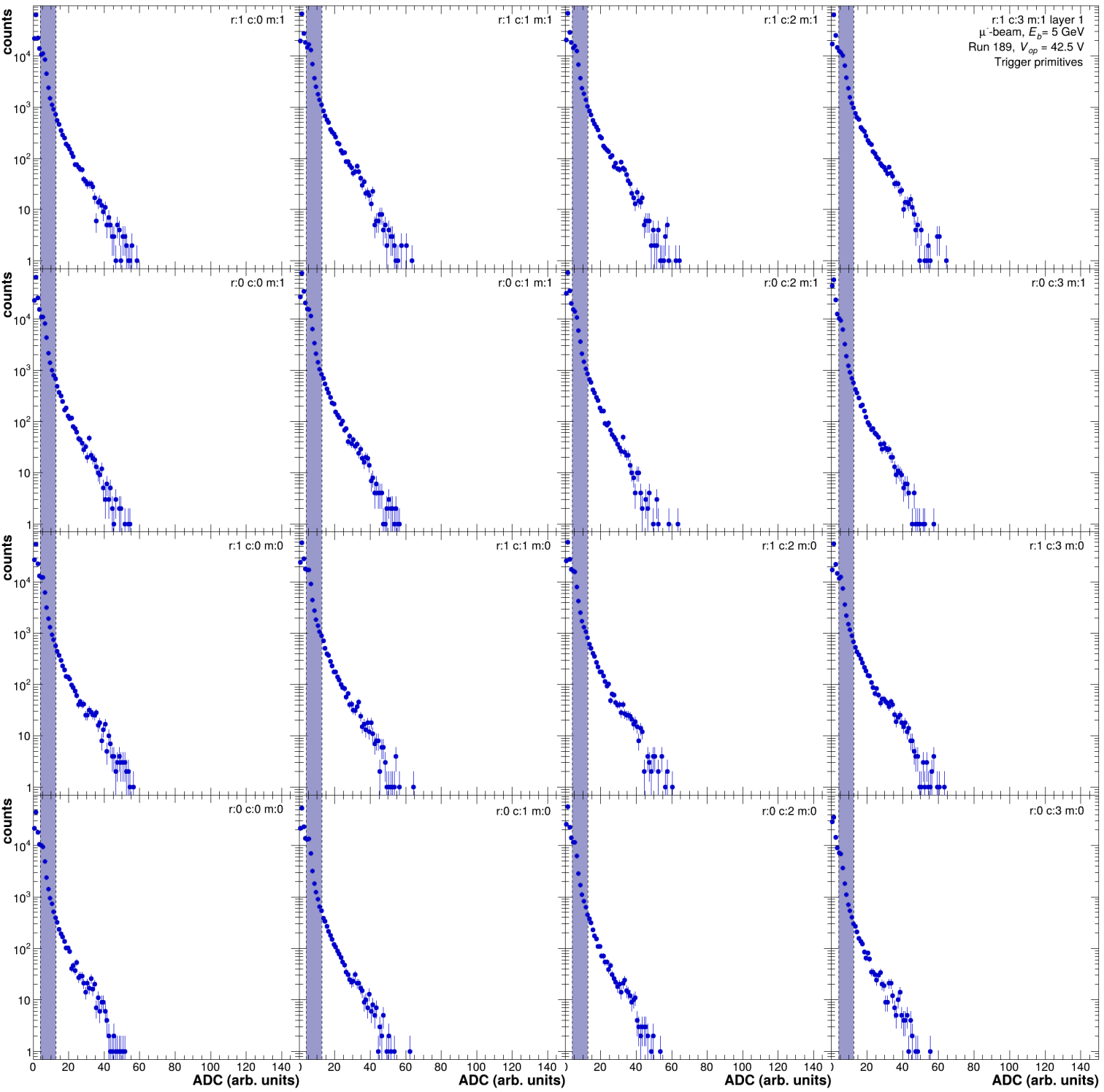

evt. collected = 799122
μ⁻-beam, E_b= 5 GeV, Run 189, V_{op} = 42.5 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
counts

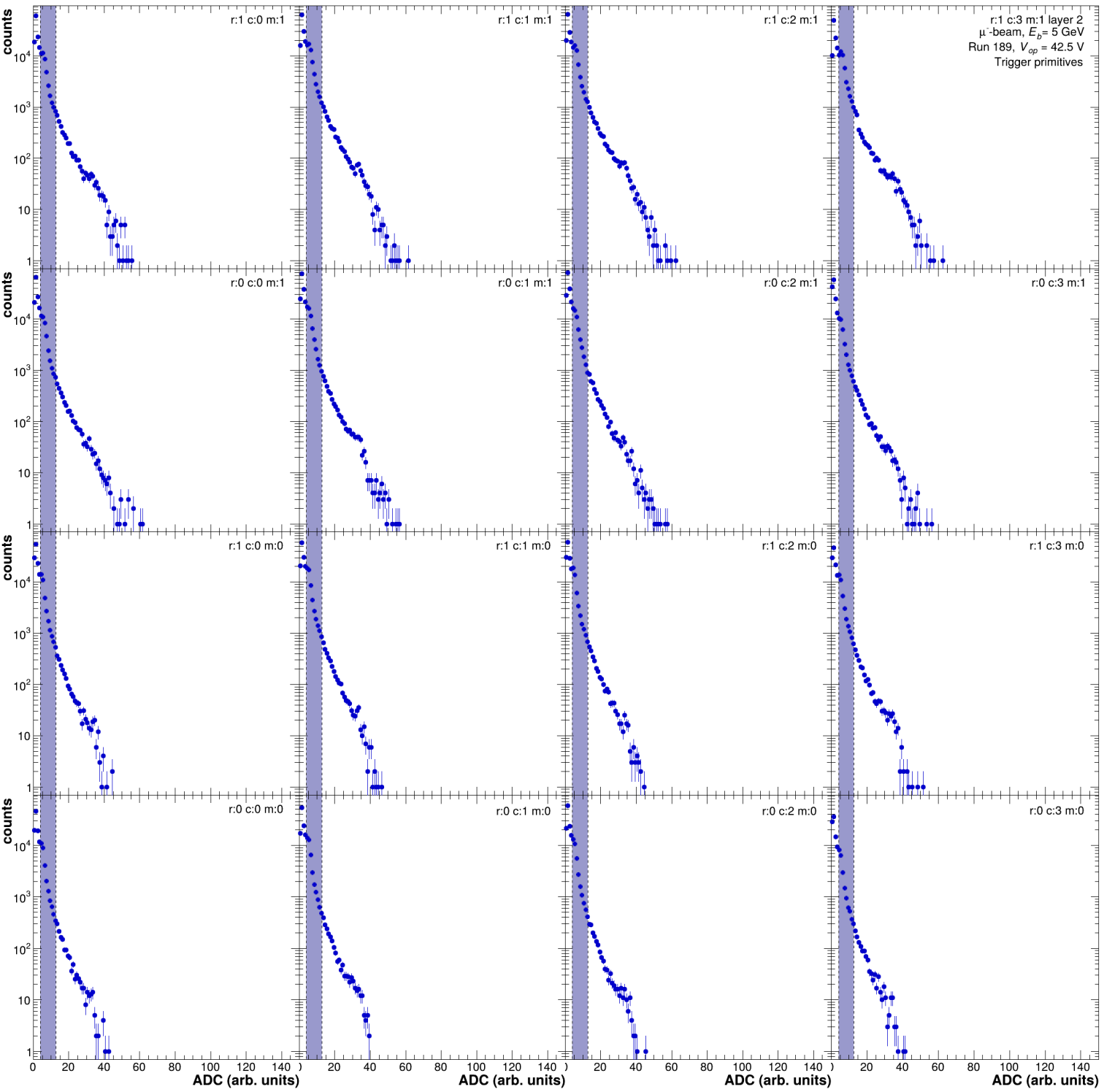

μ⁻-beam, E_b= 5 GeV, Run 189, V_{op} = 42.5 V
brd channel
layers
SNR
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
2
4
6
8
10
12
14
16
18
20
22
24

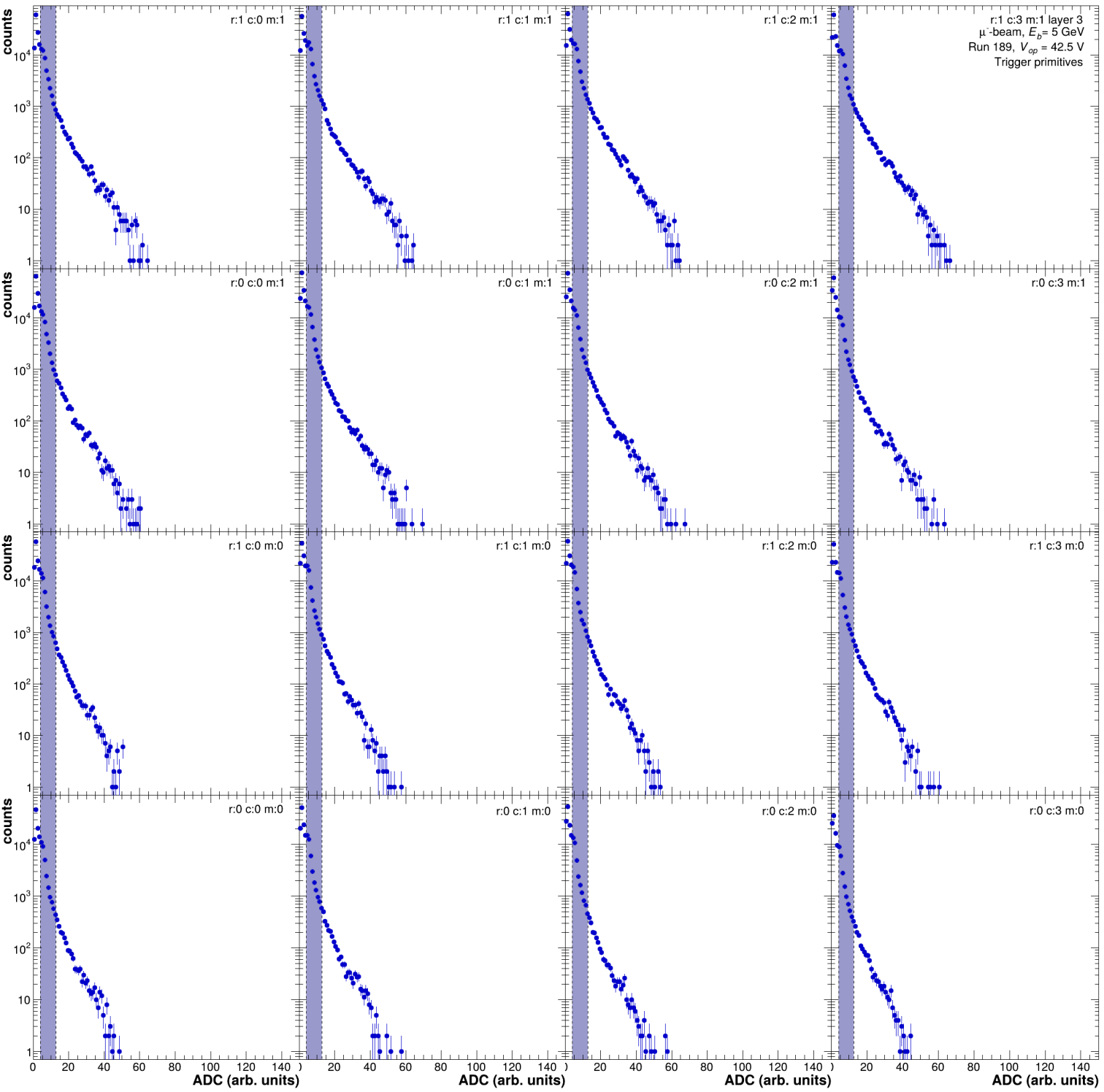

⟨ S/B noise ⟩ = 0.029
μ⁻-beam, E_b= 5 GeV, Run 189, V_{op} = 42.5 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B noise region
10^{-1}
10^{-2}

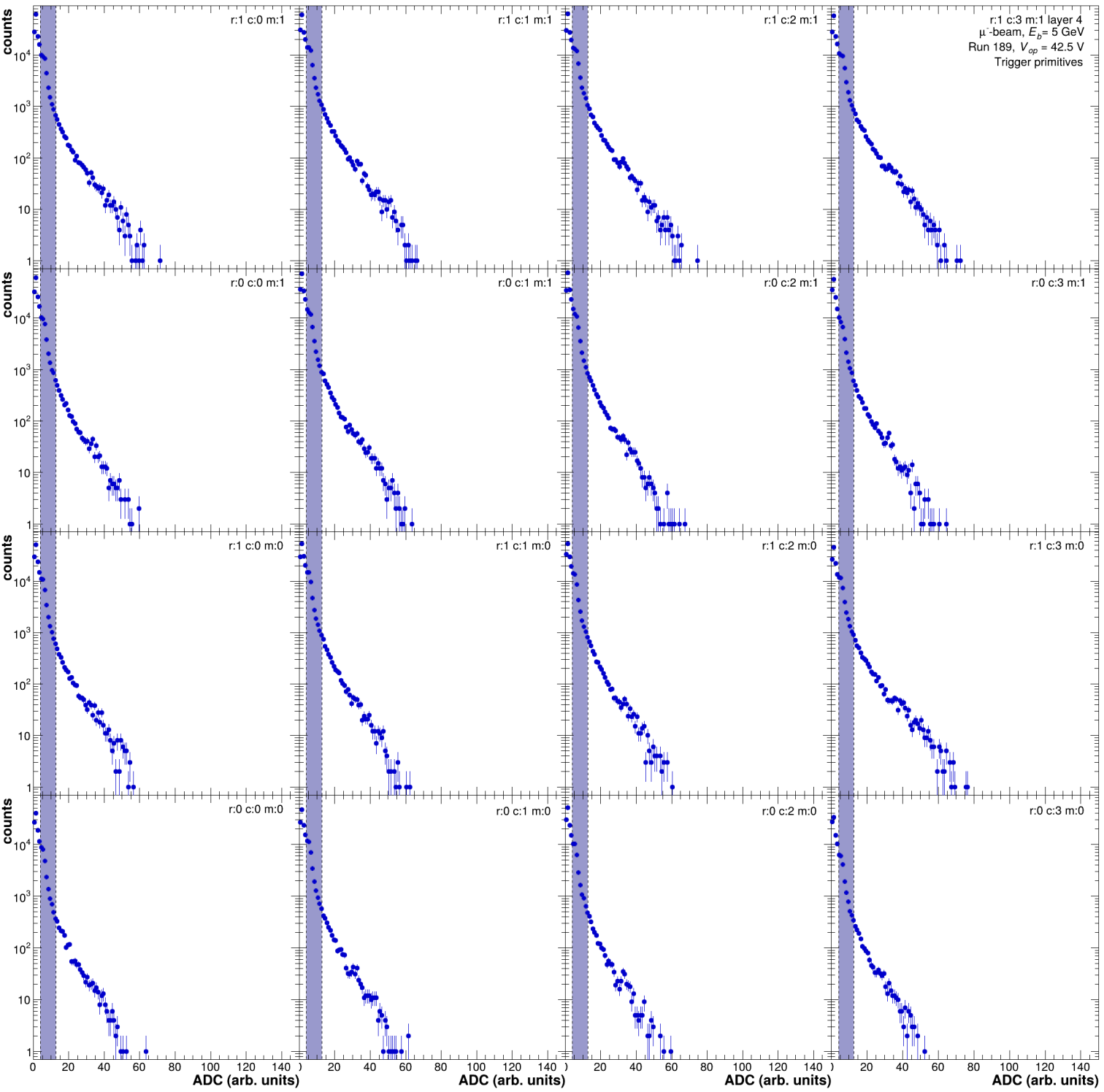

⟨ S/B signal ⟩ = 0.233
μ⁻-beam, E_b= 5 GeV, Run 189, V_{op} = 42.5 V
brd channel
layer
S/B signal region
10^{-1}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

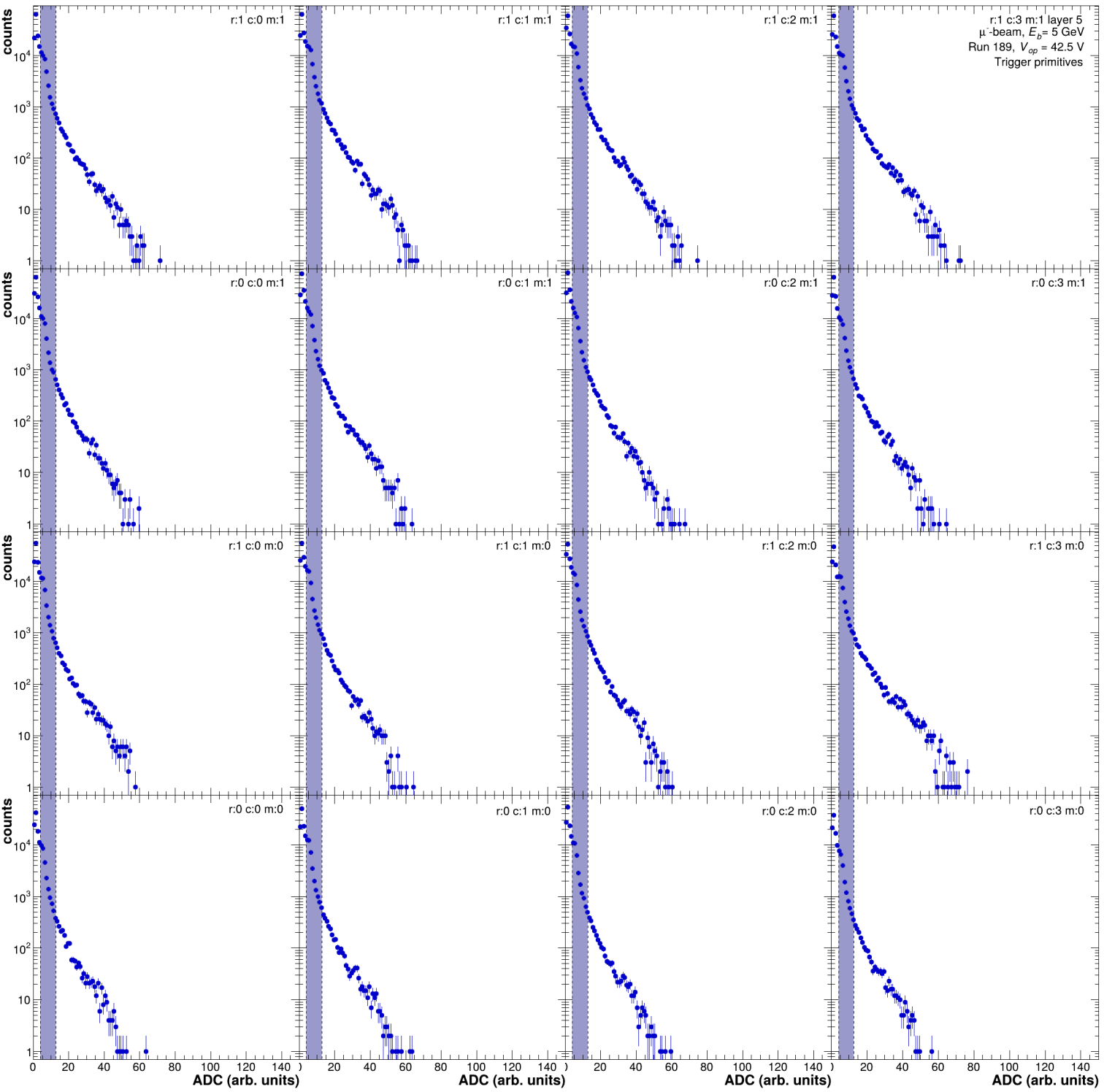

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 189, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

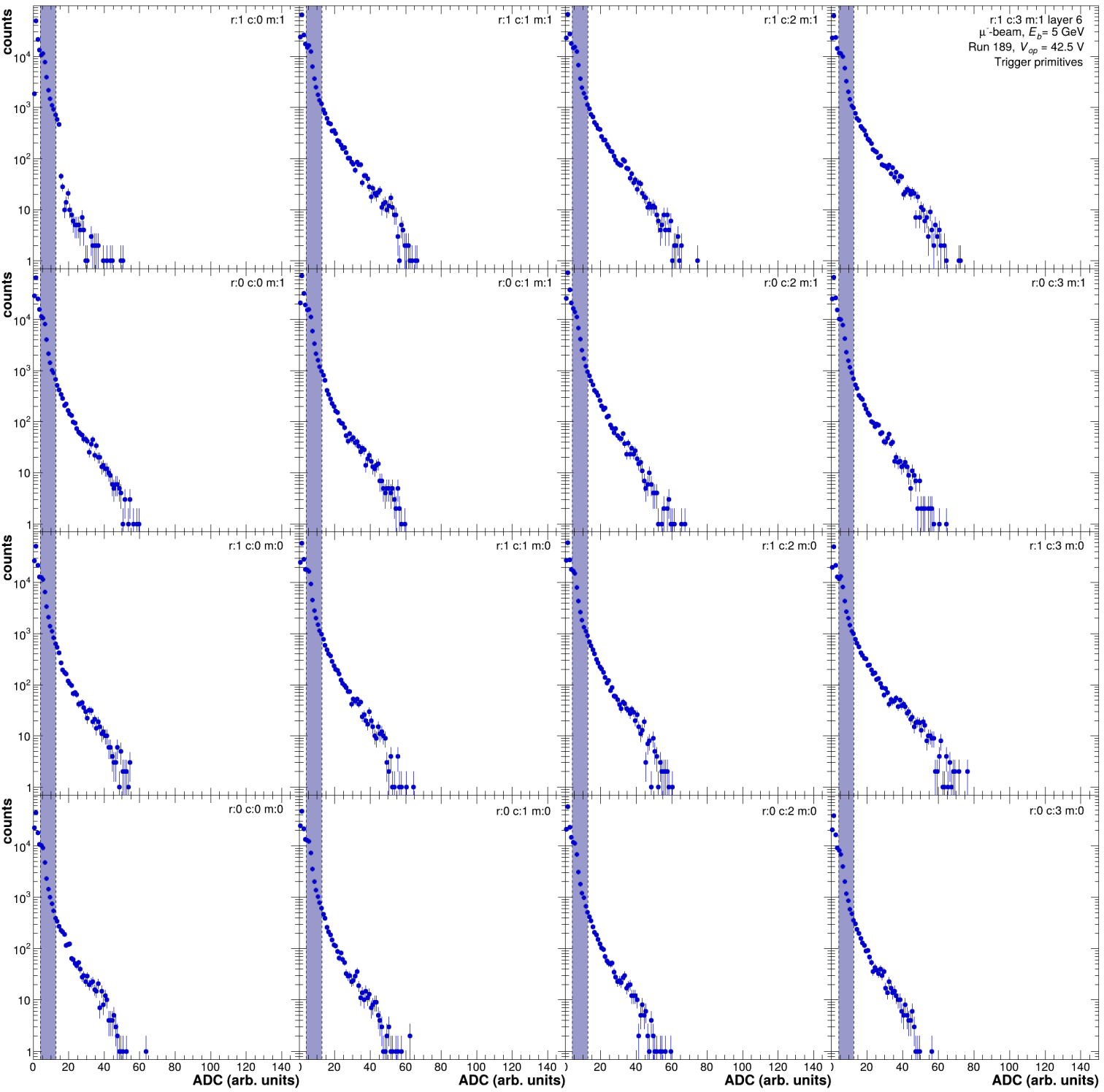

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 189, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

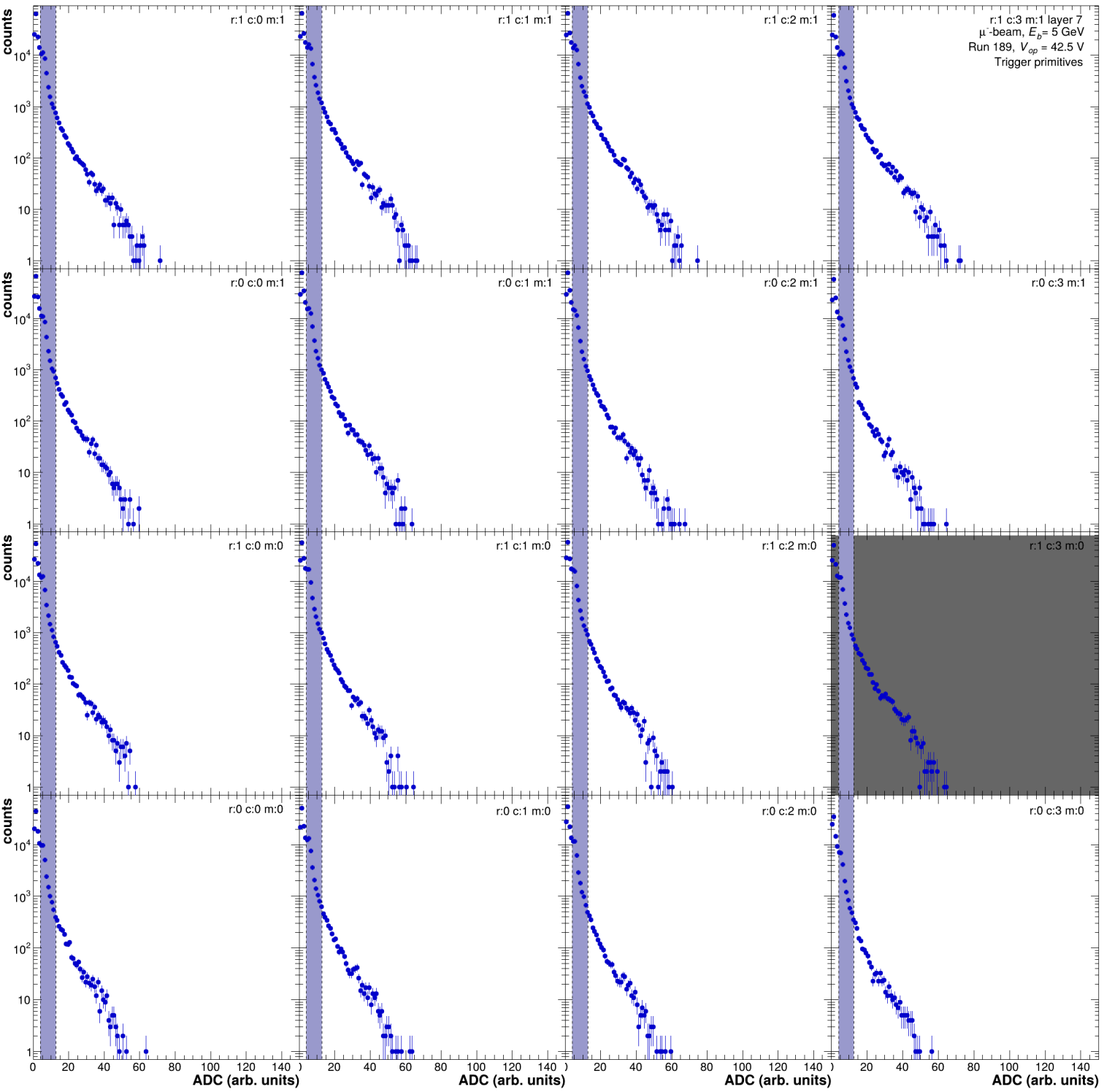

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 189, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 189, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 189, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 189, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 189, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 189, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)